

Social, creators & messaging

LUCKNOW, SUNDAY 9:15 PM, THE FAMILY WHATSAPP GROUP

Farhan's family group has 47 members and has sent 212 messages today: good-morning flowers, a cousin's wedding card, a forwarded video that claims to be "breaking news", and three voice notes from his aunt. At 9:15 pm his cousin Sana posts a link. "WhatsApp is going to charge for groups. Everyone join our Telegram group instead." Twenty-two people join within an hour. By Wednesday the Telegram group is silent. The wedding plans, the flowers and the voice notes are all back on WhatsApp.

Try it: why did the move fail, and what would it take for a rival to pull a family group away? Think for five minutes before reading on.

How a free app pays for itself

Social apps and messaging apps are free to use. They earn in other ways. A feed app such as Instagram sells your attention to advertisers, a few seconds at a time. A messaging app such as WhatsApp can't put ads inside your private chats without losing your trust, so it charges businesses that want to talk to you. Everything a PM does in this sector comes back to three questions: will people keep coming back, will enough of them create things worth watching, and can the app earn from that without making it worse?

2 bn people use Instagram every day (July 2026); India is its largest market, with about 40 crore users	50 crore+ WhatsApp users in India (2026 estimates), more than in any other country	\$60.8 bn Meta's revenue in one quarter, April–June 2026, up 28%; almost all of it from ads	≈ 12× what Meta earned per US or Canadian user compared with an Asia-Pacific user (last regional split, Q4 2023)
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Who does what, and who pays whom

Five groups meet on every social app. **Users** scroll, react and message; they pay nothing, but they give their time and their data. **Creators** post the videos and photos that users watch; a few earn a living, most post for fun. **Advertisers**, from Amazon to a bakery in Kochi, pay the platform to show ads to the right people. **Businesses** pay WhatsApp for each message they send to customers. And the **platform** decides what each person sees, runs the ad auctions and keeps the servers going. Figure 5.1 follows the money.

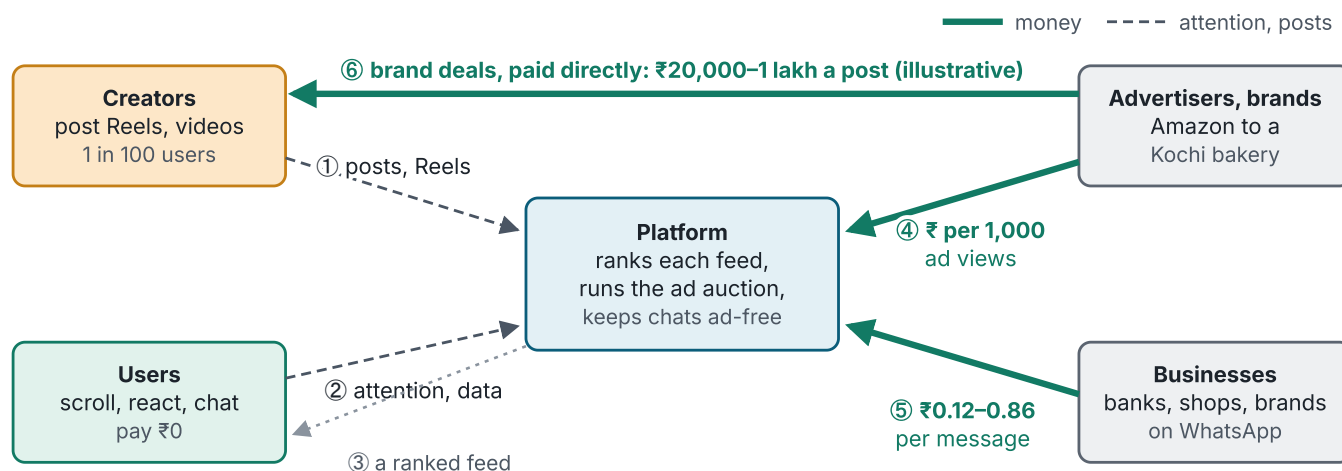
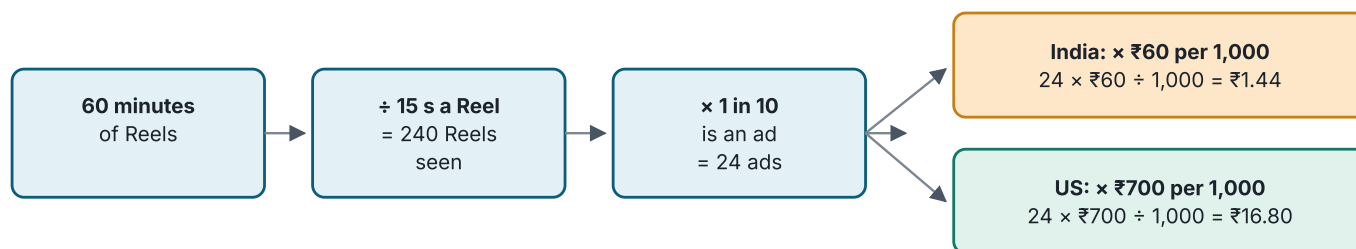


Figure 5.1 · Who pays whom on a social app. Users pay with attention, not money. Advertisers pay the platform; brands also pay creators directly for sponsored posts, often bypassing the platform. Messaging earns from businesses instead of ads in chats. Author's diagram; WhatsApp India rates from Meta's business pricing, January 2026 (approximate).

The unit economics, worked in rupees

Take one hour of Reels. A Reel lasts about 15 seconds, so an hour of scrolling shows about 240 of them. Say 1 in 10 is an ad: **24 ads**. Advertisers pay per 1,000 views, the **CPM** (cost per mille, the price of 1,000 ad views). In India a Reels CPM might be around ₹60; in the US, around ₹700 (both illustrative). So one hour earns $24 \times ₹60 \div 1,000 = ₹1.44$ from an Indian user and $24 \times ₹700 \div 1,000 = ₹16.80$ from an American one (Figure 5.2). Same app, same hour, about twelve times the money.



Same app, same hour, same 24 ads: about 12× the money. India wins on users, not on revenue per user.

Figure 5.2 • What one hour of scrolling earns. The number of ads is the same; the price per view is not. Indian advertisers pay less because Indian shoppers spend less online, so each view is worth less to them. *Illustrative CPMs and ad load; the ratio matches Meta's last published regional revenue per user (Q4 2023).*

That gap shows up in Meta's own numbers. The last time it split revenue per user by region, a US or Canadian user earned it about \$68 in a quarter, a European \$23, and an Asia-Pacific user under \$6 (Figure 5.3). This is **ARPU** (average revenue per user: revenue ÷ users, for a period). India is a **volume market**: huge numbers of users, each worth little. It shapes every product decision. An Indian feature must be cheap to run, light on data and able to work on a ₹8,000 phone.

Revenue per Facebook user, October–December 2023

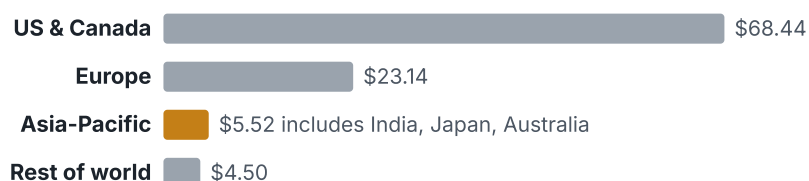


Figure 5.3 • Revenue per user, by region, one quarter. An Asia-Pacific user brought in a twelfth of a North American user. Meta stopped publishing this split after 2023; the gap has not closed. *Meta Q4 2023 results (Facebook app users, quarterly ARPU, rounded).*

Why a free messaging app is hard to earn from

Ads inside a chat with your mother feel like someone reading over your shoulder. So WhatsApp keeps chats ad-free and charges **businesses** instead. Since 2025 it bills per message, by type. In India, from January 2026, a **marketing** message costs a business about ₹0.86, a **utility** message (an order update, a bill) about ₹0.115, and replies to a customer who wrote first are free. A clothing brand that sends 2 lakh order updates and 50,000 offers in a month pays $2,00,000 \times ₹0.115 + 50,000 \times ₹0.86 = ₹23,000 + ₹43,000 = ₹66,000$. Meta's "other revenue", mostly this business messaging, reached **\$1 bn** in April–June 2026, up 73%.

The metrics that matter

Metric	What it means, in one line	Why a PM watches it
DAU and MAU	People who open the app in a day; in a month	Size. Watch which one a company reports: a daily number is harder to earn
Stickiness (DAU ÷ MAU)	Share of monthly users who come on a given day	WhatsApp is near 80–90%; a new app at 20% is visited once a week
Time spent	Minutes per user per day	Drives ad views; too much invites "addictive" criticism and rules for teens
Sends and shares	Posts shared into private chats, per user	Instagram's favourite signal: a share brings a friend back too
Creators posting	Accounts that post at least once a week	Supply: without new posts the feed gets stale
Ad load, CPM	Share of posts that are ads; price per 1,000 ad views	Revenue = views × ad load × CPM; more ads can mean less scrolling
ARPU	Revenue ÷ users, per quarter	Shows where money comes from; India low, US high
Reports per million views	How often users flag spam, abuse or scams	Trust: the guardrail on almost every growth idea

Players and numbers

Latest reported users, 2025–26 (blue = Meta; green = Meta, counted daily)

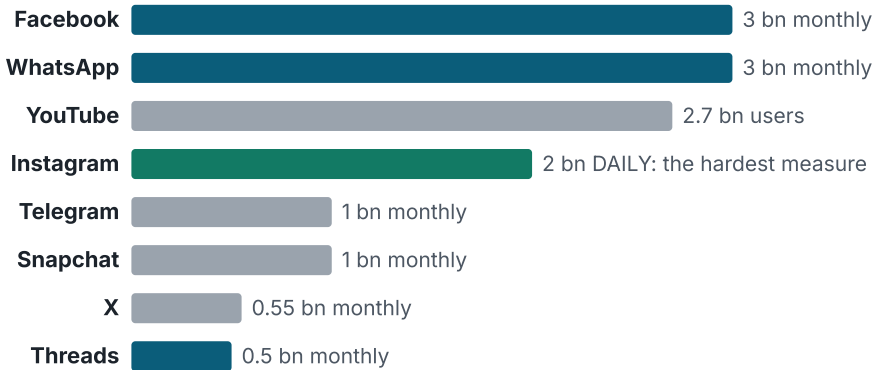


Figure 5.4 • How big each network is. Watch the measure: Instagram reports daily users, the others monthly users, which flatters them. Meta owns three of the four biggest. *Meta Q2 2026 (Instagram, Facebook, WhatsApp, Threads); Alphabet (YouTube, 2026); Snap Q2 2026; X audited figure in SpaceX S-1 (2026); Telegram (2025).*

Player	What it is	Latest numbers (dated)
Instagram	Photos, Stories, Reels and DMs; India's favourite social app	2 bn daily users (Jul 2026); India about 40 crore+
Facebook	The original network; groups, Marketplace, video	3 bn+ monthly users; Meta, all apps: 3.60 bn daily (Q2 2026)
WhatsApp	Private messaging, groups, Channels, business chat, UPI payments	3 bn+ monthly users; India 50 crore+ (2026 est.)
YouTube (Shorts)	Long videos plus short vertical ones	2.7 bn users; India about 52 crore (2026)
Snapchat	Camera-first chat with close friends	~1 bn monthly users; +10 mn daily users in Q2 2026, mostly outside the US and Europe
X	Public posts, news and argument; owned inside SpaceX's group	550 mn monthly users; ads about \$2 bn in 2025, half of 2022
Telegram	Cloud messaging, channels with no member limit, bots	1 bn+ monthly users; India its biggest download market (2025)
ShareChat • Moj	Indian-language social and short video	About 15 crore monthly users together (2026)

Four turning points

Each turning point moved hundreds of millions of users, or changed what platforms must do for them.



Figure 5.5 • How Indian social media changed in ten years. Cheap data brought hundreds of millions online; a ban and a privacy scare moved them between apps; then the rules arrived. *Company announcements; MeitY (TikTok ban, 29 June 2020; IT Rules, February 2021; DPDP Rules, November 2025); Telegram founder's post (January 2021).*

1. Cheap data (2016). Reliance Jio launched in September 2016 with free data and then very low prices. Mobile data became cheap enough to watch video all day, and the next 30 crore Indians came online through WhatsApp and YouTube, often in Hindi and other languages, often on their first smartphone. **2. The TikTok ban (June 2020).** India blocked TikTok and 58 other Chinese apps. About 20 crore short-video users were suddenly free. Moj, Josh and MX TakaTak raced for them; Instagram launched Reels in India in July 2020, within two weeks of the ban, and won. **3. The privacy scare (January 2021).** WhatsApp asked users to accept a new policy about sharing data with Facebook. Rumours spread that it would read chats. Worldwide, Telegram gained 25 million users in 72 hours, India giving it more installs than any other country, and Signal was downloaded 75 lakh times in five days. WhatsApp delayed the policy and explained that chats stayed encrypted. Most people stayed. **4. The rules arrive (2021–27).** The IT Rules 2021 made large platforms appoint grievance officers, remove flagged content quickly and, for messaging apps, be able to trace who first sent a message. The Digital Personal Data Protection Rules, notified in November 2025, bring consent and data-security duties from about May 2027.

Five mechanics that aren't obvious

1. Most people watch; few post. Roughly 90% of users only watch, 9% react or comment now and then, and 1% create most of what everyone sees (Figure 5.6). This **90-9-1 rule** is a rough pattern, not a law, but it explains why apps pamper creators: lose a few thousand top creators and the feed goes stale for millions.

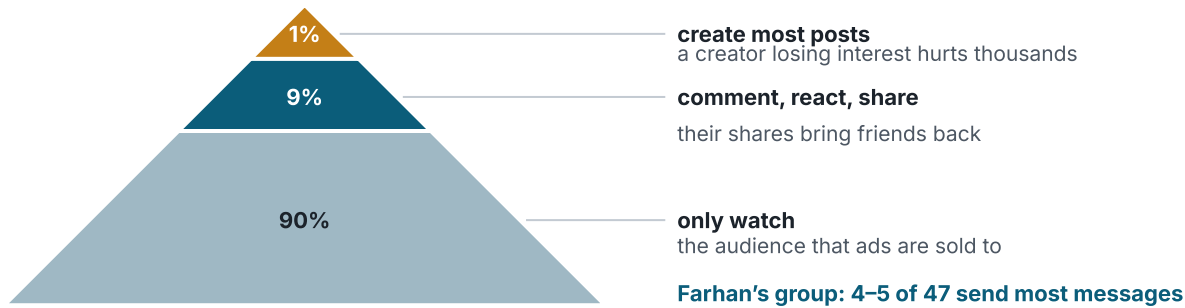


Figure 5.6 · Who makes what everyone watches. A small group of creators supplies most of the posts. Features that help one creator post more can matter more than features for a million viewers. *The 90-9-1 pattern (Nielsen Norman Group, 2006), a rule of thumb; exact shares differ by app.*

2. A ranking system decides who sees your post. Nobody sees everything. For each person, the app scores thousands of candidate posts by predicting what they will do: watch to the end, share, comment, or tap “not interested”. Each prediction gets a weight, and the post with the highest total goes first (Figure 5.7). Weights are product decisions. When Instagram said DM shares matter most, creators started making posts people want to send to a friend.

Reel A, for Priya, 22, Pune. The app predicts each action, multiplies by a weight, and adds up.

	chance she does it	× weight	= points	
Watch to the end	0.40	× 1	+0.40	
Like	0.10	× 1	+0.10	
Comment	0.03	× 5	+0.15	
Share to a friend	0.02	× 20	+0.40	
“Not interested”	0.01	× -30	-0.30	
Score for Reel A			0.75	Reel B scores 0.60, so A goes first

Figure 5.7 · How a feed scores one Reel for one person. A share is rare but weighted heavily; a “not interested” wipes out much of the score. Change a weight and you change what millions of people watch. *Illustrative probabilities and weights; the structure follows Meta’s published descriptions of feed ranking (2023–25).*

3. Friends to strangers. Feeds once showed posts from people you follow (the **social graph**). TikTok showed that posts picked from anyone, by what you watch (the **interest graph**), hold attention far better, and a new user needs no friends for the app to work on day one. Instagram and Facebook rebuilt their feeds this way from 2022. The price is that the app feels less personal, so both keep adding features for friends. **4. Messaging spreads rumours faster than feeds.** A forward travels through trusted people and nobody can see the whole chain, not even the company when chats are encrypted. WhatsApp’s answer was design, not deletion: from 2019 a message can be forwarded to at most five chats, and since April 2020 a “forwarded many times” message to only one chat at a time. **5. Brands pay creators directly.** A sponsored Reel can earn a mid-sized Indian creator ₹20,000–1 lakh (illustrative) without the platform seeing a rupee. That is why platforms build tools for tagging products and brand deals: to get a share of money that already flows around them.

IN THE INTERVIEW · QUESTIONS THIS SECTOR LOVES

- **“How does Instagram make money if it’s free?”** Users × time × ad load × price per 1,000 views; better ranking raises the price because ads work better.
- **“Should WhatsApp show ads in chats?”** No: trust is the product. Earn from businesses, Status and Channels instead, which is what Meta chose.
- **“Design a new social app for X.”** Start with the smallest group that works on its own (one college, one trade in one area), not with “everyone in India”.
- **“Engagement fell 5%. Diagnose.”** Split creators from viewers first: fewer posts or less watching? Then new vs old users, and any ranking change.
- **“How would you fight misinformation?”** Limits on forwards, labels and fact-checks, without breaking encryption.

CASE 5.1 · LINKEDIN FOR BLUE-COLLAR WORKERS

Product design

Medium IIM B Sigma PM Casebook 2026–27, pp. 46–47

Try it first. Think of the last time your family hired a cook, a driver or someone to clean. How did you find them, and what did you check before letting them into the house? Now think about it from the worker's side. Spend five minutes on how an app could help both.

THE QUESTION

Let's design something like LinkedIn, but for blue-collar workers in India. How would you approach this?

- A** Before I jump in, can I clarify two small things? First, are we focusing only on India? Second, do you want the platform to help with full-time jobs, part-time gigs, or both?
- Q** Good questions. Yes, India. And let's include both full-time and part-time work.
- A** Great, that helps. Let me begin with the overall context then. Blue-collar workers in India rely mostly on word-of-mouth and local contacts. There's no proper digital identity for them. Employers also don't have one reliable place to find verified workers. So, the overall context is a huge workforce, high churn, very little structure, and very low trust.
- Q** Makes sense. How do you understand the users?
- A** Let me create two simple personas, so we keep real people in mind. The first one is Ramesh. He's 28, a delivery worker. He changes jobs every few months because timings or salary change suddenly. He hates dealing with agents because they charge a fee. The second one is Savita. She's 35 and works as a housekeeper. She wants work near her home because she has kids to look after. She's not very confident reading long English text, so she prefers simple visuals and audio guidance. For employers, imagine someone like Mr. Shah who runs a small restaurant. He always needs staff but spends too much time asking other restaurants or agents. And if someone leaves suddenly, his whole kitchen slows down.
- Q** Nice. What pain points do you see here?
- A** Ramesh's biggest pain is job uncertainty. He doesn't know when he'll get his next job or how far he'll need to travel. Savita's pain is trust, she worries about safety, late payments, or unexpected changes in work hours. For employers like Mr. Shah, the pain is unreliability, people not showing up, wrong identity details, and high churn. So overall, the big pains are unclear job information, trust issues, distance problems, and no proper worker profile.
- Q** Good observation. Now how would you prioritize the problems to solve?
- A** I would start with the simplest problems that unblock everything else. First priority: job discovery. If workers can't see clear, nearby jobs, the whole platform fails.
Second: identity and trust. If employers don't trust the profiles, they won't hire through us.
Third: clarity -salary, timing, weekly off, working hours. Most disputes happen because these aren't fixed upfront. Other things like training, community features, or financial services can come later.
- Q** Alright. What direction does this set for your solution?
- A** It means the platform should focus on giving every worker a simple digital profile, showing nearby verified jobs with transparent pay, and supporting basic identity checks. Almost like creating "mini LinkedIn pages" for people who've never had one.
- Q** And what choices do you need to think about?
- A** The biggest choice is whether this should be a new app or part of an existing one. Given the audience, I'd build a new app with a very simple flow, big buttons, audio instructions, multiple languages. Another choice is where to start. I would start with delivery workers, drivers, and housekeeping staff because they already use smartphones and switch jobs quickly. I also need to choose how deep verification needs to go. I think basic phone and Aadhaar verification is enough for MVP, with optional deeper checks later.
- Q** Good. How would you actually execute the MVP?
- A** The first step would be a very easy sign-up for workers. They enter their phone number, pick their language, take a photo, and choose their job type using icons. They can also record a short voice or video intro if they want, something simple like "I'm Ramesh, I have 3 years of delivery experience." The app then shows nearby jobs with salary and timings clearly visible. Ramesh just taps "Apply," and the employer sees his profile. On the employer side, they add their location, choose the job type, mention salary and timings, and the app shows matching workers. Once both sides agree, the app creates a small digital agreement, so expectations are clear.
- Q** Nice flow. How does this tie back to success metrics?

A Success should be measured in simple, real-world terms. First, how many workers complete their profile? That tells us if onboarding is easy enough. Second, how many job matches actually convert into someone starting work? That's our core metric. Third, what's the 30-day retention of both workers and employers? If people keep coming back, it means trust is building. And finally, distance between worker and job, shorter distances show the matching algorithm is working.

Q Nice. That's a clear and very grounded approach. Thank you for the breakdown.

A Thank you

WHAT WORKED · WHAT WAS MISSING

- ✓ **Three personas, including the employer.** A jobs app has two sides, and Mr. Shah's pain (people not showing up) is as real as Ramesh's.
- ✓ **A clear order of problems:** discovery, then trust, then clarity of pay and hours, each with a one-line reason.
- ✓ **A flow built for the user:** icons, voice intro, local language, jobs near home, a simple written agreement.
- ✓ **The right core metric:** matches that turn into someone actually starting work.
- ✗ **Never asked what exists today.** Apna has about 6 crore job-seekers in 900+ cities; WorkIndia and local agents are everywhere; e-Shram has registered 31.89 crore workers. Why would Ramesh switch?
- ✗ **No plan for the empty app.** Which city, which trade and how many employers must be on the app before Savita sees enough jobs to come back?
- ✗ **Scams were missing.** Fake job offers that ask for a "registration fee" are one of India's most common frauds. The employer must be checked, not only the worker.
- ✗ **Metrics count profiles, not hires that last,** and there is no guardrail (wage disputes, no-shows, safety complaints) and no answer to who pays.

Scoping kit check: Confirm the thing ✓ **India** · Goal ✗ · Who ✓ **three personas** · Today and the trigger ✗ · Edges ✓ **full- and part-time**. Design add-ons: platform ✓ **new app** · eligibility and legal limits ✗. Skipping "what exists today" cost the most: the answer never says why a worker already on Apna would try one more app.

CHECK · "BASIC PHONE AND AADHAAR VERIFICATION IS ENOUGH FOR THE MVP" **OVERSTATED**

A phone OTP is easy. Aadhaar is not: private apps can't freely run Aadhaar authentication, so most use documents the user chooses to share through DigiLocker, with consent. The bigger gap is direction. Scams usually run against the worker: a fake "company" asks for ₹500–2,000 to "confirm the job" and disappears. So the employer needs checking at least as much as the worker, and the app should promise one simple rule: **a worker never pays to get a job.**

STRONGER ANSWER



Figure 5.8 · Six moves for a two-sided product. Start with the smallest area where the app works, before designing screens. *Author's summary.*

A ① CLARIFY

Is this LinkedIn itself, with its brand, or a new company? Is the goal hires, or engagement? And what do workers use today? I know Apna, WorkIndia, local agents who charge a month's salary, and word of mouth. I'll assume the goal is **hires that last**, in a new app under the LinkedIn brand.

Q That's fine. LinkedIn wants a foothold with India's blue-collar workforce. Hires are the goal.

Why this move: a jobs app that grows "profiles" but not hires is a directory nobody returns to. Naming Apna and the agents early forces the answer to say why anyone would switch.

A ② START SMALL

I won't launch across India. I'll pick one trade in one area where both sides are close together: home help (cooks, cleaners, nannies) in gated societies of one Pune suburb, say Wakad. Savita comes back only if she sees at least 3 suitable jobs within 3 km each week. About 1 in 3 jobs fits her hours, so she needs $3 \times 3 = 9$ new postings a week nearby. If 2% of households post a job in a given week, we need $9 \div 2\% = 450$ **active households** (Figure 5.9). Ten societies of 500 flats, with 1 in 10 signing up, gives 500. That is reachable in a month through the societies' committees.

Why this move: an empty app loses both sides on day one. The maths turns "get users" into one target the team can hit: 450 households in one suburb.

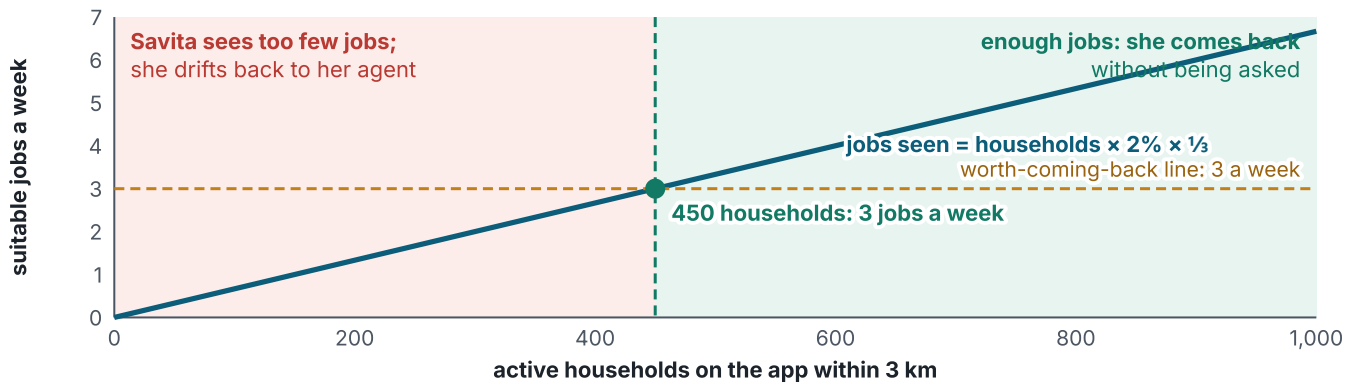


Figure 5.9 • When does the app start to work for Savita? Below about 450 active households nearby she sees fewer than 3 suitable jobs a week and drifts away; above it, she comes back without being asked. *Illustrative: 2% of households post in a week; 1 in 3 jobs fits a worker's hours.*

A ③ TRUST, BOTH WAYS

Households fear theft; workers fear unpaid wages and fake jobs. So: employers verify their phone and flat through the society; workers verify their phone and share an ID through DigiLocker; the app helps with police verification, which many cities expect for home help. Pay, hours and weekly off are fixed in a one-page agreement before the first day. And the rule “a worker never pays” is on the first screen, so any request for money is a red flag (Figure 5.10).

Why this move: trust is the product here. Each check blocks one known way things go wrong, and the “never pay” rule turns every worker into a scam detector.

The scam

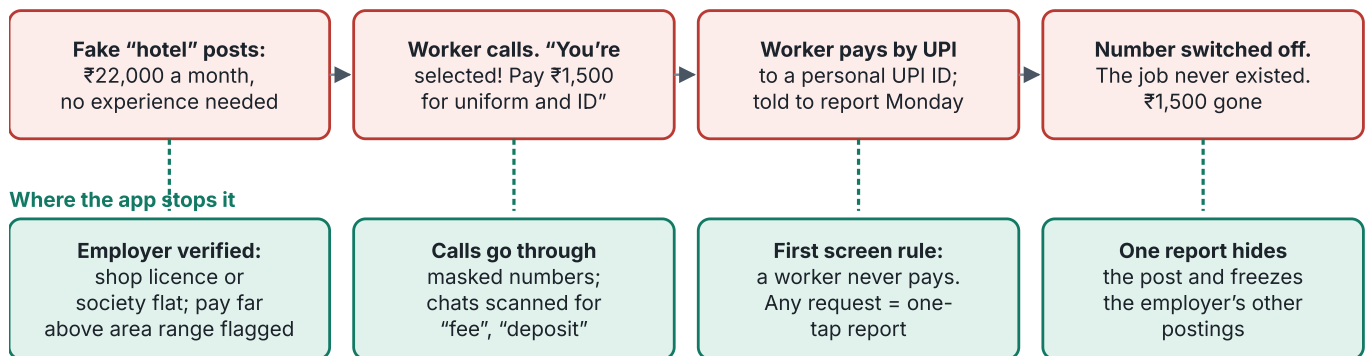


Figure 5.10 • A fake-job scam, and where the app stops it. Each checkpoint breaks one step of the fraud. The cheapest one is a rule every worker can remember. *Scam pattern from NCRB's cyber-awareness material on job fraud; checkpoints are the author's design.*

A ④ FLOW

For Savita: pick Marathi or Hindi, record a 20-second voice intro, tap icons for the work she does and the hours she is free. She sees jobs on a map with pay and hours in big numbers, and taps “Call” through a masked number, so her phone number stays private. A trial day, then the agreement. On the 1st of each month, the household gets a reminder to pay her wage by UPI.

Why this move: each screen removes one reason Savita gives up: reading English, typing, distance, sharing her number with strangers, or chasing her pay.

A ⑤ WHO PAYS

Households pay, workers never do. An agent in Pune often charges a month's salary, ₹8,000–12,000. We charge ₹499 per hire, refunded if the worker leaves within 15 days. Illustrative first quarter: 500 households × 30% hire = 150 hires × ₹499 ≈ ₹75,000. That money is small; the point of the first suburb is to prove the density maths, then copy it society by society.

Why this move: charging the side that has money and a problem to solve keeps the app free for the side we need most, and the refund makes the promise believable.

A ⑥ MEASURE

Goal (North Star): hires still working after 30 days, per week. **Inputs:** suitable jobs seen per worker per week (target 3+); share of postings answered within 24 hours; trial-to-hire rate. **Safety checks (guardrails):** wage disputes and scam reports per 100 hires; no-shows. **Trade-off:** strict checks slow down new employers, but loose checks bring scams, and one scam story in a society WhatsApp group can empty the app. **Next:** when a suburb passes 3 jobs a week per worker, move to the next suburb, then add drivers.

Why this move: “still working after 30 days” counts only hires that helped both sides, which a count of profiles or matches can't tell you.

HOW THE ANSWER HANGS TOGETHER

The goal, hires that last, set the North Star. The density maths (3 jobs a week within 3 km, so 450 households) chose where to start. Trust checks and the “never pay” rule protect the goal from its biggest threat, and the scam guardrail watches exactly that. Money comes from the side with the problem, and expansion waits until the first suburb works.

FRAMEWORK · NETWORK EFFECTS AND THE COLD START

1 · The situation. Savita downloads the app, sees one job 9 km away, and never opens it again. Mr. Shah posts a job, gets no replies in two days, and goes back to his agent. The app works only when both sides are there, close together, at the same time.

2 · An everyday example. A new weekly haat (street market) in a colony. Buyers come only if there are enough stalls; vendors come only if there are enough buyers. A smart organiser doesn't advertise across the city. She picks one street, one day (Sunday), persuades 20 vendors with free stalls for a month, and tells three nearby colonies. Once Sunday on that street is worth the walk, it fills itself.

Idea	The Sunday haat	The jobs app
Two sides	Vendors and buyers	Workers and households
Smallest network that works	One street, one day	One trade in ten societies of one suburb
"Worth coming back"	Enough stalls to buy the week's vegetables	3 suitable jobs a week within 3 km
Seed the hard side	Free stalls for the first month	Field team signs up households through society committees
Tipping	It becomes "the" Sunday market	It becomes where Wakad hires home help
Expand	Next street, next day	Next suburb, then the next trade

3 · The framework. A **network effect** means each new user makes the product more useful for the others. There are three kinds (Figure 5.11). **Direct:** WhatsApp is useful because your people are on it. **Cross-side:** more households make the app better for workers, and more workers make it better for households. **Local:** only users nearby count; a cook in Wakad gains nothing from a household in Delhi. The **cold start** is the empty-app problem: with too few users, nobody finds value and nobody stays. The cure is the **atomic network** (Andrew Chen's term): the smallest group that works on its own. Build one, then copy it.

How fast does value grow? A group of n people has $n \times (n - 1) \div 2$ possible pairs. Farhan's 47-member family group has $47 \times 46 \div 2 = 1,081$ pairs. But nobody talks to all 46 relatives; people talk to a handful. So value grows with users, but much more slowly than the count of pairs suggests, and a bigger group can even get worse: more noise, more forwards, more muting.

Using it, step by step (on this case):

1. Name the sides and which one is harder to get: households who post jobs.
2. Define "works" as one number a user feels: 3 suitable jobs a week within 3 km.
3. Find the smallest area and trade where that number is reachable: 450 households in one suburb.
4. Seed the hard side by hand: society committees, a field team, a first month free.
5. Expand only when the first network passes the number, and copy the playbook.

COMMON SLIP · spending on national advertising at launch. Thin everywhere means empty everywhere: a lakh sign-ups spread across India can still leave every worker seeing zero jobs nearby.

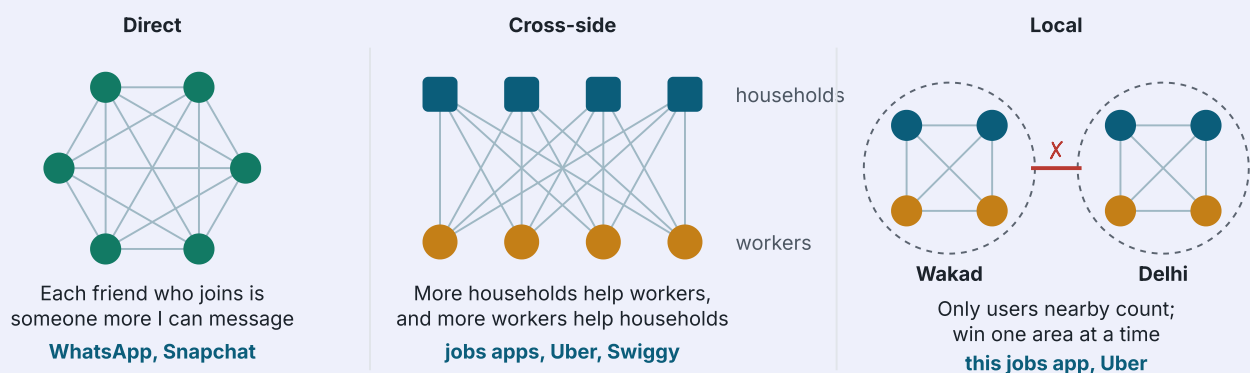


Figure 5.11 · Three kinds of network effect. A jobs app has the cross-side kind, and it is also local, so it must be won one area at a time. *Author's diagram.*

4 · When it misleads. A network effect protects you only if users can't easily use a rival too. Workers keep Apna and yours on the same phone, and take whichever job comes first, so the advantage must be earned every week. Big numbers can also hide an empty product: Google+ got crores of sign-ups through Gmail in 2011, but people's friends weren't posting there, so it stayed a ghost town and shut in 2019. And a network can turn bad as it grows: once scammers find a jobs app, every new fake posting makes it worse for everyone.

5 · In the interview. "This only works if both sides are close together, so I'll start with one trade in one suburb, define the density that makes a worker come back, and hit it before expanding."

Try it first. You bought a kurta after a creator's Reel. List everyone who gained from that sale: you, the creator, the brand, Instagram. Now ask which of those gains Instagram can actually see and measure. Five minutes.

THE QUESTION

You are a product manager at Instagram working on the Creator Commerce product tagging feature. How would you go about measuring its success?

A Before I begin, let me confirm my understanding. Creator Commerce allows creators to tag products from partner brands in their posts, reels, and stories. Users can tap the tag, view product details, and complete the purchase on the merchant's website. Is that accurate?

Q The description is accurate. Please proceed with the rest of the solution.

A Great. Since this feature serves multiple stakeholders, I'd first map their journeys before defining metrics.

The three stakeholders are:

- Creator
- Buyer
- Brand partner

If that's okay, I'd first cover the creator and buyer journeys.

Q The approach seems good. Let's keep the focus on the creators and buyers. You can skip the brand-side metrics for now. Please go on with the journeys.

A Sure.

Creator Journey:

- Discover the Creator Commerce program.
- Verify eligibility and enable product tagging.
- Tag products in posts, reels and stories.
- Track performance through Instagram Insights.
- Optimize content based on results.

Buyer Journey:

- Discover tagged products while browsing Instagram.
- Open the product detail page.
- Visit the merchant website.
- Complete the purchase.
- Return to Instagram for future engagement.

Q You've captured the journeys accurately. Please go ahead with the metrics.

A I'll define metrics separately for creators and buyers.

Creator Metrics:

Awareness:

- % eligible creators aware of the program
- Invite CTR

Engagement:

- % creators enabling product tagging
- Tagged posts/reels per creator
- Weekly active tagging creators

Conversion:

- Outbound clicks to merchant sites
- GMV attributed to creator content
- % creators driving purchases

Q That's a solid set. What about metrics for buyers?

A I'd use a similar funnel.

Buyer Metrics:

Awareness:

- % users exposed to tagged products
- Product tag CTR

Engagement:

- Product detail views
- Time spent on product pages

Conversion:

- Merchant website CTR
- Purchase conversion rate
- Repeat outbound clicks

Q Do you think we can consider any other metrics to track for this program?

A Yes. I'd also monitor operational and guardrail metrics

Operational Metrics:

- Outbound link success rate
- Merchant page load time
- Attribution accuracy

Guardrail Metrics:

- Creator satisfaction
- Abuse/fraud reports
- Buyer complaints about checkout or product quality

Q Makes sense. How would you go about defining a North Star and tracking all these metrics?

A Since Instagram doesn't own checkout, the North Star should reflect successful commerce driven by creators.

North Star : Creator-attributed Purchase GMV (or Outbound Purchase Conversion from Creator Content)

To measure this, we'd instrument events across the funnel:

- Product tag impressions
- Product tag clicks
- Product detail views
- Outbound merchant clicks
- Purchases attributed using creator and campaign IDs

I'd also capture creator and buyer feedback through lightweight post-purchase surveys.

Q Thanks for the great discussion. We can end the interview.

WHAT WORKED · WHAT WAS MISSING

- ✓ **Confirmed what the feature does** before measuring it, including the detail that checkout happens on the brand's website.
- ✓ **Mapped each side's journey first**, so the metrics follow real steps rather than a list from memory.
- ✓ **Added operational metrics:** link success, page load and attribution accuracy. Few candidates think of these, and they decide whether the numbers can be trusted.
- ✗ **Never asked why Instagram wants this.** Instagram earns from ads, not from these sales. Is the goal to keep creators earning (so they post here, not on YouTube), or to win brand ad budgets? The North Star depends on it.
- ✗ **About 25 metrics, none ranked.** The interviewer can't tell which three matter, or what would make the team stop or expand.
- ✗ **GMV as the North Star rewards the wrong things:** a few expensive items or discount spam can move it while most creators earn nothing.
- ✗ **No guardrail on the feed itself.** If tagged Reels feel like ads, people scroll past, watch time falls and creators lose followers. None of the guardrails watches that.

Scoping kit check: Confirm the thing ✓ · Goal ✗ · Who ✓ **creators, buyers, brands** · Today and the trigger ✗ **new or six months old?** · Edges ✗ **which country?** Metrics add-ons: the job for each side ✓ · product stage ✗ · what decision the metric will drive ✗. Without the goal and the decision, the answer can list metrics but can't choose among them.

CHECK · "NORTH STAR: CREATOR-ATTRIBUTED PURCHASE GMV" **OVERSTATED**

GMV (gross merchandise value, the total value of goods sold) is easy to report, but three things weaken it here.

Instagram doesn't see most purchases: they happen on the brand's site, and since Apple's 2021 tracking rules many iPhone purchases can't be linked back to the tap. A single ₹40,000 watch counts as much as 40 kurtas from 40 different creators. And Instagram doesn't earn from GMV; it earns when brands spend more on ads and when creators keep posting. GMV is a useful input, not the goal.

CHECK · "PURCHASES ATTRIBUTED USING CREATOR AND CAMPAIGN IDS" **DIRECTIONALLY RIGHT**

Tagging the link with the creator's ID tells you who got the *last click*. It misses people who saw the Reel, didn't tap, and bought later by searching the brand, and it credits sales that would have happened anyway. The fair test is a **holdout**: hide tags from a random 5% of users and compare purchases from both groups.



Figure 5.12 · Six moves for a metrics question with several stakeholders. Start with the goal and the decision; the metrics follow. *Author's summary.*

A ① CLARIFY

How long has the feature been live, and where? Why does Instagram want it: to keep creators earning here, to win brand ad budgets, or both? And what decision will these metrics feed, for example whether to open tagging to every creator?

Q Live in India for six months with 20,000 invited creators. Leadership must decide whether to open it to all creators. The aim is creators earning on Instagram.

Why this move: "measure success" means nothing until you know which decision it serves. Here it is one yes-or-no call, so the answer must end with a rule for that call.

A ② GOAL

Then success is creators earning money from Instagram without making the feed worse. Each side wants something different (Figure 5.13): creators want sales, buyers want useful finds without feeling sold to, brands want sales at a fair cost, and Instagram wants creators to keep posting and viewers to keep watching.

Why this move: writing each side's win in one line shows where they clash. That clash is what the guardrails must watch.

Product choice	Creator	Viewer (buyer)	Brand	Instagram
Tags on every Reel	+ more chances to earn	- feed feels like a shop	+ more exposure	- watch time may fall
Tagged Reels shown to non-followers	+ new buyers	± useful finds, or strangers' ads	+ more reach	± good only if relevant
Higher commission	+ earns more	0 no change	- pays more per sale	+ creators stay
Rank tagged Reels higher	+ more sales	- worse feed	+ more sales	- trust falls, then ad money

Figure 5.13 · Who wins and who loses from four product choices. No choice is good for every side. The feed-quality losses on the right are what a GMV-only answer never sees. *Author's analysis.*

A ③ ONE NUMBER

Goal (North Star): creators who earned from at least one tag-driven sale this week. It grows only if many creators succeed, not a few big ones, and a creator who earns keeps posting. I'd report GMV and commission beside it, not instead of it.

Why this move: a count of creators earning rewards breadth. It moves when a small creator in Indore makes her first ₹200, which is exactly the behaviour Instagram wants to spread.

A ④ INPUTS

Follow one tagged Reel with 10 lakh views (Figure 5.14). 2% tap the tag: 20,000. 40% of them view the product: 8,000. 30% go to the brand's site: 2,400. 3% buy: 72 orders × ₹1,200 = ₹86,400 of sales, and at 8% commission the creator earns about ₹6,900 (illustrative). The biggest lever Instagram controls is the first step, the tap rate, which depends on whether the product fits the Reel.

Why this move: the chain shows where the numbers are small and which steps Instagram controls. The last step happens on the brand's site, so it is watched but can't be the main lever.

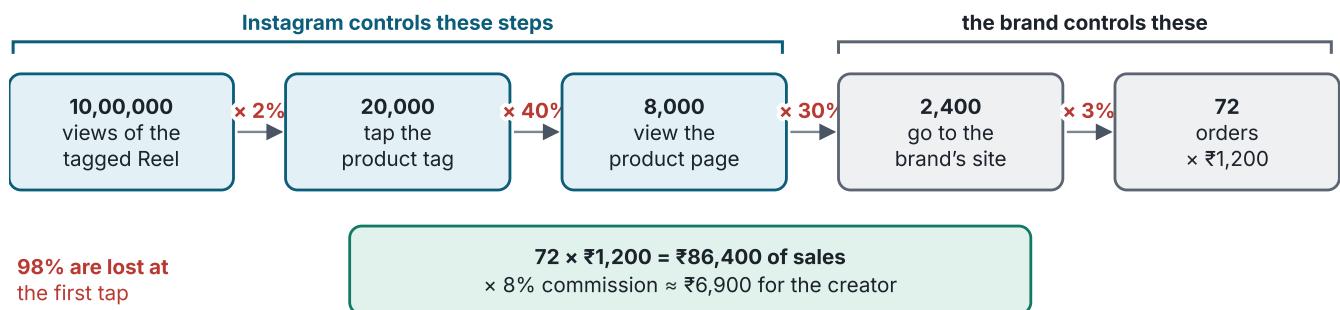


Figure 5.14 · From 10 lakh views to ₹6,900 for the creator. Most people are lost at the first tap. Instagram owns the first three steps; the brand owns the last two. *Illustrative rates; commission levels vary by brand and category.*

A ⑤ GUARDRAILS

Safety checks (guardrails): watch time on a creator's tagged Reels against the same creator's untagged ones; "not interested" taps and unfollows after tagged posts; returns and complaints about tagged products. If tagged Reels are watched 10% less, viewers are treating them as ads.

Why this move: comparing a creator with herself removes the difference between popular and unpopular creators, so any gap is caused by the tag.

A ⑥ DECIDE

Before opening it to everyone, run a holdout: hide tags from 5% of viewers for four weeks. Open to all creators if (a) creators earning grows week on week, (b) tagged Reels' watch time stays within 3% of untagged ones, and (c) complaints per 1,000 orders stay flat. **Trade-off:** more tags earn creators more but make the feed feel like a shop. **Next:** let brands boost a creator's tagged Reel as an ad, which is where Instagram earns.

Why this move: the rule turns the metrics into the yes-or-no call leadership asked for, and the holdout proves the sales are extra rather than relabelled.

HOW THE ANSWER HANGS TOGETHER

The decision (open to all creators?) and the goal (creators earning) chose the North Star: creators with at least one sale a week. The funnel showed that the tap rate is the lever Instagram controls. The guardrails compare each creator with herself so feed damage shows up clearly, and the holdout plus a three-part rule turns the numbers into a decision.

KEY IDEA · THE WHO-WINS, WHO-LOSES GRID

List the product choices down the side and every stakeholder across the top, then mark each cell with a plus or a minus. It takes two minutes on a whiteboard. It shows the interviewer that you have thought about everyone the feature touches, and every minus becomes a guardrail.

CASE 5.3 · USERS MOVING FROM WHATSAPP TO TELEGRAM

Strategy

Medium

IIM B Sigma PM Casebook 2026–27, pp. 149–150

Try it first. Open your phone. Which groups do you have on Telegram and which on WhatsApp? Why is each one where it is? If WhatsApp disappeared tomorrow, which of your groups would move, and which would simply die? Five minutes.

THE QUESTION

How to stop people moving from WhatsApp to Telegram?

A I have some initial clarifying questions. First, What are the primary reasons users are switching to Telegram, unique features Telegram has, and any recent triggers (past 6 months) causing the migration, including trust factors and user experience issues?

Q Users are switching due to recent updates by Telegram focused on privacy and user control

A What is WhatsApp's main goal in addressing this issue? Is it user retention, feature parity, or maintaining market dominance?

Q You are free to decide.

A I believe the main goal should be user retention through feature enhancement and improved user experience, as this will naturally lead to maintaining market dominance. Before proposing specific solutions, I'd like to identify different user segments and their pain points. Is that acceptable?

Q Sure, please proceed.

A I can identify 4 potential customer segments –

1. Privacy-conscious users: Concerned about data security and encryption.
2. Power users: Require advanced features and higher file-sharing limits.
3. Group communication users: Extensively use groups and community features.
4. WhatsApp Business users: Businesses leveraging WhatsApp for customer communication

I'd like to focus on Privacy-conscious users and Power users as the primary segments, as addressing their needs will likely have the most significant impact on preventing migration to Telegram. If that's agreeable, I can proceed with listing pain points?

Q Yes, that is a reasonable approach. You can proceed.

A Pain points for Privacy-conscious users:

1. Concerns about data sharing with Facebook (now Meta)
2. Lack of anonymity options
3. Limited control over personal data
4. Concerns about government surveillance features.

Pain points for Power users:

1. File size limitations for sharing media and documents

2. Lack of advanced customization options
3. Limited multi-device support
4. Absence of features like message scheduling and self-destructing messages

The following can be the proposed features to address these pain points:

1. **Enhanced Encryption:** Implement end-to-end encryption for all types of communication, including backups, with user-controlled encryption keys.
2. **Anonymous Chat Mode:** Introduce an option for users to chat without sharing their phone numbers, using temporary usernames instead.
3. **Increased File Sharing Limits:** Raise the file size limit for sharing media and documents to match or exceed Telegram's limits.
4. **Advanced Customization:** Provide more options for app themes, chat backgrounds, and notification settings.

Q How will you prioritize among these?

A For prioritizing features, I will consider their potential impact on user trust and implementation complexity:

Feature	Impact	Complexity
Enhanced Encryption	High	High
Anonymous Chat Mode	High	Medium
Increased File sharing limits	Medium	Low
Advanced Communication	Medium	Medium

Q How will you measure the success of these features implemented?

- A**
1. **User Retention Rate:** Measure the percentage of users who remain active on WhatsApp over time to assess the effectiveness of the new features
 2. **Feature Adoption Rate:** Track the usage of newly introduced features to determine user engagement and acceptance.
 3. **Churn Rate:** Monitor the rate at which users are leaving WhatsApp for Telegram

Q What would be the North Star metric among these?

A User Retention Rate is a direct measure of the percentage of users who remain active on WhatsApp over time. This metric provides a clear indication of how effectively WhatsApp is maintaining its user base and preventing churn. By focusing on retention, WhatsApp can ensure long-term user engagement and loyalty, which are critical for sustaining growth and countering the shift of users to Telegram. High retention rates typically reflect positive user experiences and satisfaction, which are essential for keeping users from switching to alternative platforms.

Q Great! We can close the case now.

A Thank you.

WHAT WORKED · WHAT WAS MISSING

- ✓ **Asked why users are leaving**, and set a goal when the interviewer left it open.
- ✓ **Split users into segments** and chose two, with a reason.
- ✓ **Linked each feature to a pain point** and ranked them by impact against effort.
- ✗ **Three of the four features already existed.** WhatsApp has encrypted every chat end to end since 2016 and offered encrypted backups with a user's own password or 64-digit key since 2021; it raised the file limit to 2 GB in 2022, the same as free Telegram. The gap was awareness, not features.
- ✗ **Never defined "moving".** Uninstalling WhatsApp, or starting new groups on Telegram while keeping WhatsApp? Almost everyone does the second.
- ✗ **Never sized it.** How many, which age group, which country?
- ✗ **Didn't test the interviewer's reason.** Telegram's normal chats are not end-to-end encrypted; only "secret chats" are. People who move for privacy often move to less of it.

Scoping kit check: Confirm the thing ✗ **what does "moving" mean?** · Goal ✓ · Who ✓ **segments** · Today and the trigger ✓ **asked why** · Edges ✗ **country, age group**. Strategy add-ons: size of the problem ✗ · existing assets ✗ · competitor ✓. The costliest skip, again, was "what exists today": a check of WhatsApp's own settings would have saved three of the four ideas.

CHECK · THE FOUR PROPOSED FEATURES AGAINST WHAT WHATSAPP ALREADY HAD DOESN'T HOLD UP

Proposed feature	What already existed	Verdict
End-to-end encryption for everything, including backups, with user-held keys	Every chat and call end to end since April 2016; optional encrypted backups with a password or 64-digit key since late 2021	Already there; the job is telling people
Anonymous chat without a phone number	Not in 2024–25; WhatsApp was reported to be building usernames	A real gap, with a spam risk
Bigger file limits	2 GB per file since 2022 (Telegram: 2 GB free, 4 GB paid)	Already equal
Customisation, scheduling, self-destructing messages	Chat themes and wallpapers; disappearing messages since 2020; up to 4 linked devices since 2021	Mostly there

STRONGER ANSWER

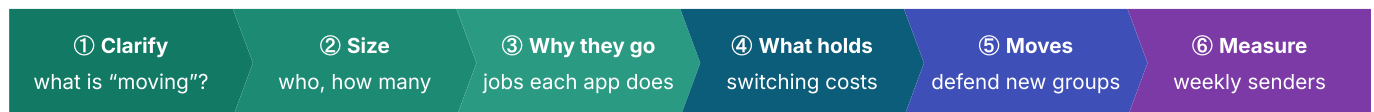


Figure 5.15 · Six moves for a “stop users leaving” question. Define leaving, size it, and learn what each app does better before proposing anything. *Author’s summary.*

- A ① CLARIFY**
What does “moving” mean in our data: people uninstalling WhatsApp, or using it less while Telegram grows? Which country and age group? I’ll assume India.
- Q** India. Daily users are flat, but among 18–24s new groups are increasingly started on Telegram, and their time on WhatsApp is slipping.
- Why this move: “leaving” turned out to mean “starting new groups elsewhere”. That is a different problem from uninstalling, and needs a different fix.*
- A ② SIZE**
Illustrative: about 10 crore WhatsApp users in India are 18–24. If 15% now start their new groups on Telegram, that is 1.5 crore people. Few will uninstall: in January 2021, the biggest scare WhatsApp ever had, Telegram gained 25 million users worldwide in three days, yet WhatsApp stayed India’s most used messaging app (Figure 5.16).
- Why this move: the 2021 episode is a natural experiment. Even a huge scare moved new activity, not whole families, which tells you where the real risk lies.*

January–May 2021: downloads spiked, but people added a second app rather than leaving.



Figure 5.16 · January 2021: the biggest scare, and why most people stayed. Millions downloaded Telegram and Signal, but family groups stayed put. WhatsApp explained, delayed the policy, and the new apps became second homes rather than replacements. *Telegram founder’s post (12 January 2021); Sensor Tower via TechRadar (Signal, Telegram installs); WhatsApp announcements (January–May 2021).*

- A ③ WHY THEY GO**
Each app does a different job. Telegram wins where people **don’t know each other**: channels with lakhs of followers (exam notes, stock tips, movie updates), groups up to 2 lakh, usernames so you needn’t share your number, and chats stored in the cloud so a new phone gets everything. WhatsApp wins where people **do know each other**: family, friends, colleagues, local shops. Privacy is rarely the real reason, since Telegram’s normal chats are less private.
- Why this move: naming the job each app does explains who moves and who doesn’t. A college batch of 400 strangers is a Telegram job; the family group is not.*
- A ④ WHAT HOLDS THEM**
Farhan’s family group shows the strongest force: every member must move together, or the group splits. If each of 47 members has a 90% chance of joining, the chance all of them do is 0.9 multiplied by itself 47 times, about **0.7%** (Figure 5.17). Add years of photos and voice notes, and a phone book that is already the contact list. So WhatsApp won’t lose its old groups. It is losing the *new* ones.
- Why this move: this separates what is safe (old groups) from what is at risk (new, large groups), so the fixes can aim at the right place.*

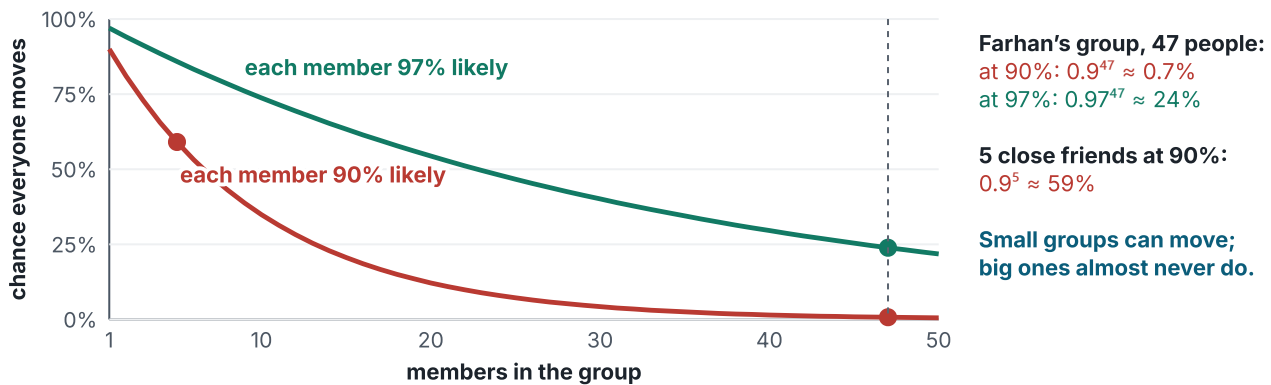


Figure 5.17 · Why whole groups almost never move. Even when each person is very likely to follow, the chance that everyone does falls fast as the group grows. One grandmother who stays is enough to keep the family where it is. *Simple probability, assuming each member decides independently (illustrative).*

A ⑤ MOVES

Three moves, in order. **1. Tell the privacy story** where people look: a one-line banner in new group chats, “Only members can read this group; not even WhatsApp”, and a nudge to switch on encrypted backup. Cost: almost nothing. **2. Usernames for large groups**, so a college batch can join without sharing phone numbers, with message requests from strangers kept in a separate folder. **3. Better communities for 18–24s**: bigger announcement groups inside Communities, and file sharing that matches Telegram. I would not copy Telegram’s open public channels for everything; they bring piracy and scams.

Why this move: the first move fixes a belief, which is cheap; the second and third close the real gaps for the one job WhatsApp is losing, without giving up what makes it trusted.

A ⑥ MEASURE

Goal (North Star): weekly active senders aged 18–24, rather than “users who remain”. **Inputs:** new groups created per week by 18–24s; share of those groups with 50+ members; encrypted-backup switch-ons. **Safety checks (guardrails):** spam and unknown-sender reports (usernames can bring strangers in); messages per family group. **Trade-off:** the more WhatsApp opens up to strangers, the less it feels like a private space. **Next:** track people who use both apps, to see which jobs keep moving.

Why this move: counting senders, not openers, catches the real warning sign: people who still open WhatsApp but start their conversations elsewhere.

HOW THE ANSWER HANGS TOGETHER

Defining “moving” turned the problem into “new groups start elsewhere”. The 2021 episode and the 0.7% maths showed that old groups are safe and new, large ones are at risk. The moves target exactly that job, and the North Star (weekly senders aged 18–24) and guardrails (spam, family-group activity) watch the same thing.

FRAMEWORK · SWITCHING COSTS, AND WHY PEOPLE KEEP BOTH APPS

1 · The situation. Telegram was free, had bigger groups and promised privacy. Farhan’s family still came back in three days. Something other than features was holding them.

2 · An everyday example. Moving house to another city. The rent is lower and the flat is bigger, yet many people stay. Moving costs money (packers, deposit), you lose things (old papers, the plants), you must learn a new place (routes, shops, a new doctor), and hardest of all, you leave your neighbours, friends and the school your children love. And many people who work in a new city keep the old house for weekends: they live in two places for different needs.

Kind of cost	Moving house	Moving a WhatsApp group
Money	Packers, deposit, brokerage	Almost nil: both apps are free
Things you lose	Old papers, furniture that won’t fit	Years of photos, voice notes, chat history
Learning	New routes, new shops	New buttons, new settings
People	Neighbours, friends, school	All 47 members must move together
Keeping both	Old house for weekends	WhatsApp for family, Telegram for channels

3 · The framework. A **switching cost** is everything a user loses, pays or must redo to change products. There are four kinds: money, data (what you’ve built up), learning, and people (the others who must move with you). For messaging the last one is by far the largest (Figure 5.18). **Multi-homing** means using two rival products for different jobs, like carrying two SIM cards, one for cheap data and one for calls. When keeping both costs almost nothing, a rival doesn’t need to win your old habits. It only needs to win one new job, and grow from there.

Using it, step by step (on this case):

1. List the four costs for one real user, here one family group, and say which is biggest: people.
2. Ask whether keeping both apps is cheap. Here it is: both are free and sit on the same phone.
3. Find the job the rival does better, for which people will multi-home: large groups of strangers.

4. Defend the new job, not the old one: old groups are held by people-costs already.
5. Raise good switching costs (history, trust), never bad ones (locking data in). Bad ones invite regulators.

COMMON SLIP · treating high switching costs as safety. They protect what users already do on your product, not what they will start next. Most losses begin as a second app for one new job.

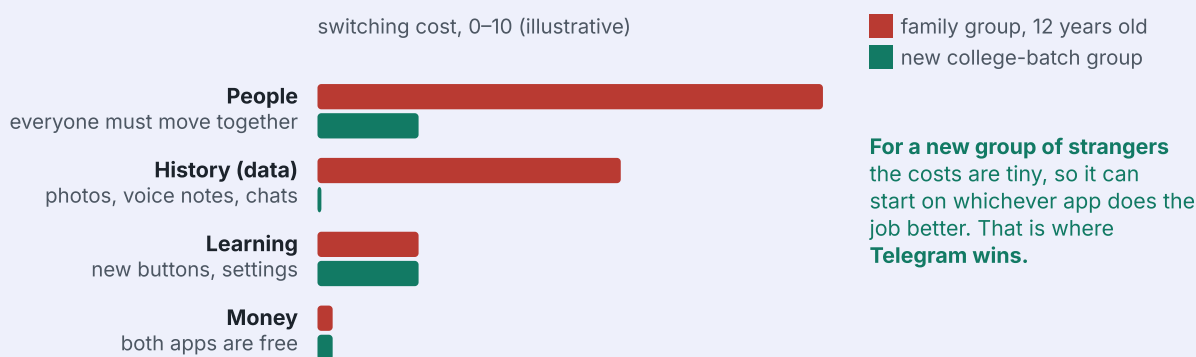


Figure 5.18 · What it would take to move Farhan's family group. Money and learning are small. The history and, above all, the 47 people are what hold the group. A rival must win through a job where those costs are low. *Author's illustrative scores for one family group.*

4 · When it misleads. Switching costs can be cut from outside. Mobile number portability (2011) let Indians change telecom operators without losing their number, and people switched far more freely afterwards. Europe's Digital Markets Act now requires large messaging apps to let users of other apps message their users, which lowers the "people" cost. Lock-in also hides unhappiness: users who stay only because leaving is hard are gone the moment it becomes easy.

5 · In the interview. "Old groups won't move because everyone would have to move together; the risk is new groups for jobs Telegram does better, so I'll defend those jobs and measure where new groups start."

SOCIAL, CREATORS & MESSAGING · GUESSTIMATES

Guesstimates: counting what people do

Social and messaging guesstimates ask how often people do something: post, send, share, call. The trap is to treat every user as the same person. A few people post daily; most never do. A teenager sends hundreds of messages; her grandfather sends five. Split users by behaviour, and anchor the answer to a published total. The six cards still do the work.

1

Pin down the unit

A post, a Reel, a Story? A chat, a message, a conversation? Per day, on a typical day?

2

Pick the approach

Usually bottom-up from users: people × share active × how often each acts.

3

Draw the tree

Population → online → on the app → active that day → split by behaviour → actions.

4

Round numbers, one reason each

Split by what people do (watchers, sometimes-posters, creators), not by age alone.

5

Check it another way

Actions per user per day, and India's share of a global figure the company has published.

6

Give a range, and what moves it

Usually the share of heavy users or the rate for the biggest group.

KEY IDEA · INDIA'S SLICE OF A GLOBAL NUMBER

Companies often publish one global number (WhatsApp: 100 billion+ messages a day; Instagram: 2 billion daily users). India's share is roughly its share of users, adjusted for how intensely Indians use the app. If India has 18% of the users and uses the app about as much as everyone else, expect about 18% of the activity, not a third. This single check catches most social-media guesstimates that have gone wrong.

GUESSTIMATE 5.1 · INSTAGRAM POSTS A DAY IN INDIA

Guesstimate

Easy IIM A Product Mindset 2026–27, p. 235

Try it first. How many posts (not Stories) did you put on Instagram last month? Your five closest friends? Is anyone you know posting every day? Build the daily number for India from that, for five minutes.

THE QUESTION

Estimate the number of Instagram uploads occurring in India.

A A few clarifying questions first. Should I estimate uploads per day or per year, and should "uploads" include posts, stories, or both?

Q Consider only posts and estimate uploads per day.

A Should I consider both personal and business accounts?

- Q** Personal accounts only for now.
- A** Can I assume this estimate is for a typical day, not a festival or a holiday?
- Q** You can assume a typical day and start working on your approach.
- A** I'll break the problem down as:
 Instagram uploads per day = active personal Instagram users × average posts per user per day.
 First, active personal users:
 active users = population × % with smartphone and internet × % on Instagram × % monthly active
 India's population is about 1.4 billion. Assuming 70% smartphone and internet penetration:
 $1.4 \text{ billion} \times 70\% = 980 \text{ million smartphone users with internet.}$
- A** Should I take Instagram penetration across all smartphone users, or segment by app usage?
- Q** What do you think?
- A** Instagram is among the most downloaded apps on the Play Store, with over a billion downloads, so I'll assume around 50% of smartphone-internet users have it.
- Q** Sure, good point. For simplicity, assume 50% penetration among smartphone users and feel free to make reasonable assumptions.
- A** Thanks, assuming 50% of these people use Instagram: $980\text{M} \times 50\% = 490\text{M}$ accounts
 I'll assume that 80% of these accounts are active personal accounts: $490\text{M} \times 80\% = 400\text{M}$
 So, we have around 400 M active personal users.
- Q** Sounds reasonable.
- A** Now, I'll estimate the average number of uploads per person per day. We can divide the monthly active users into three categories, with rough estimates for the percentage of users in each category from my own experience on the app as follows:
- 1. Light users:** 60% of users, posting one post every month = $0.6 \times 400 = 240 \text{ M users}$
 - 2. Moderate users:** 35% of users, posting one post every week = $0.35 \times 400 = 140\text{M users}$
 - 3. Heavy users:** 5% of users, posting one post every 2 days = $0.5 \times 400 = 20\text{M users}$
- Therefore, the number of posts per day = $240\text{M} \times (1/30) + 140\text{M} \times (1/7) + 20\text{M} \times (1/2) = 8\text{M} + 20\text{M} + 10\text{M} = 38 \text{ M}$
- Q** That sounds good. Anything else? I think Instagram has around 100 million daily posts.
- A** Then India contributing around 38 million close to a third of the global total is in a plausible range, given India's large user base and the fact that Instagram is blocked in China.
- Q** Great, thank you. We can close this interview.

WHAT WORKED · WHAT WAS MISSING

- ✓ **Pinned down the unit in three questions:** posts only, per day, personal accounts, a typical day.
- ✓ **Wrote the equation before the numbers,** then filled it step by step.
- ✓ **Split users by how often they post,** which is exactly the right split for this question.
- ✗ **A slip in the maths:** heavy users are 5%, so $0.05 \times 400 = 20$ million; the written " 0.5×400 " would be 200 million. The result used 20, so only the working is wrong, but interviewers notice.
- ✗ **The posting rates are high.** 60% of users posting every month and 35% every week makes the average Indian user post every 10 days. Most Instagram users post a few times a year.
- ✗ **The check was read the wrong way round.** India with a third of global posts would mean Indians post about twice as often as everyone else. That should have been a warning, not a confirmation.

Guesstimate kit check: 1 Unit ✓ · 2 Approach ✓ **bottom-up by behaviour** · 3 Tree ✓ · 4 Numbers with reasons ✗ **"from my own experience"** · 5 Check ✗ **done, but misread** · 6 Range ✗. Card 5 needs two checks, and one of them is per person: $38 \text{ million} \div 400 \text{ million users} = \text{one post every } 10\frac{1}{2} \text{ days per user, which is too often.}$

CHECK · "INDIA CONTRIBUTING AROUND A THIRD OF THE GLOBAL TOTAL IS PLAUSIBLE" **DOESN'T HOLD UP**

India has about 40 crore Instagram users out of roughly 3 billion monthly users worldwide: about 13%. Unless Indians post about $2\frac{1}{2}$ times as often as everyone else, India should contribute something near 13% of global posts, not 38%. And the interviewer's "100 million posts a day" is an old number from the photo era, before Reels; treat it as a rough anchor, not a fact.



Figure 5.19 · The six cards, applied to counting posts. The behaviour split (card 4) and the per-person check (card 5) do most of the work. *Author's summary.*

A 1 · UNIT

Feed posts and Reels from personal accounts in India on a typical day. Stories are out; carousels count as one post.

Why this move: "uploads" could include Stories, which are several times more common than posts. Saying what counts protects the answer from a 5× misunderstanding.

A 2 · APPROACH

Bottom-up from users: monthly active personal accounts, split by how often each group posts, then add up.

Why this move: posting is very uneven, so one average for everyone would hide the few who post most.

A 3-4 · TREE AND NUMBERS

140 crore people × 70% online = 98 crore. About 40 crore of them are active on Instagram each month. Split them by behaviour: 70% mostly watch and post about once every two months (28 crore × 1 ÷ 60 ≈ 47 lakh a day); 25% post about every 10 days (10 crore × 0.1 = 1 crore a day); 5% are keen posters and small creators, posting every other day (2 crore × 0.5 = 1 crore a day). Total ≈ 47 lakh + 1 crore + 1 crore ≈ **2.5 crore posts a day** (Figure 5.20).

Why this move: each group gets a rate with a reason you can defend: a watcher posts for birthdays and trips, a regular for weekends, a creator on a schedule.

A 5 · CHECK

Per person: 2.5 crore ÷ 40 crore users = 0.06 posts a day, one post every 16 days. That sounds right for a mix of watchers and posters. Against the world: if the rest of the world also posts about once every 16–20 days, India's 13% of users gives about 13–15% of posts. Both checks agree.

Why this move: one check per side, the person and the world, catches most mistakes. This is where the casebook answer would have been corrected.

A 6 · RANGE

About **1.5 to 4 crore posts a day**. The swing factor is the keen posters: at 3% the total falls to about 2 crore; at 8% it rises past 3 crore. With Stories added, the number would be several times higher.

Why this move: naming the swing factor tells the interviewer which assumption to argue with.

HOW THE ANSWER HANGS TOGETHER

The unit (posts and Reels, not Stories) fixed what to count. Splitting users by behaviour turned one shaky average into three believable rates, and the per-person and India-share checks both pointed to about 2.5 crore. The range names the one group, keen posters, that moves the answer most.

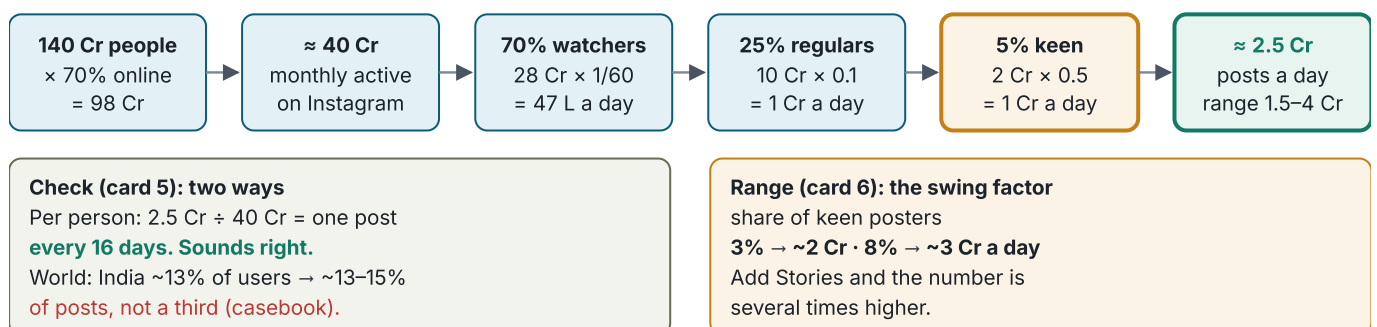


Figure 5.20 · Instagram posts a day in India, as an equation tree. The orange box is the swing factor; the green box is the answer; the grey box is the check that rejects "a third of the world's posts". *Population and online share rounded (2026); Instagram users from 2026 estimates; posting rates illustrative.*

THE REUSABLE TRICK

Actions a day = active users × share in each behaviour group × how often that group acts. Split by behaviour, then check the answer per person (does one post every 16 days sound like people you know?) and as India's share of a global number.

Try it first. Scroll your WhatsApp chat list. How many different chats, people or groups, did you send at least one message in yesterday? Is that typical? Build India's daily total from there, for five minutes.

THE QUESTION

Estimate the number of WhatsApp chats occurring in India in a day?

- A** I have some initial clarifying questions. We are focusing on chats and not on messages? Also chats includes individual chats, group chats or both?
- Q** Yes, we are focusing on the number of chats and it covers both individual and group chats.
- A** I would like to start by categorizing the Indian population based on urban and rural segments. Then, I will estimate the number of WhatsApp users in each segment. Finally, I will estimate the average number of chats per day for each group, considering both individual and group chats. Is this approach alright?
- Q** Sounds like a comprehensive approach, please proceed.
- A** Population of India is approximately 1.4 billion people. Considering the urban and rural split to be 40% and 60%. The urban and the rural population will be around 560 and 840 million. Now, assuming smartphone penetration rates in urban and rural India to be 80% and 60%, we get 448 million urban smartphone users and 504 million rural smartphone users. Does this seem reasonable?
- Q** Yes, you may continue.
- A** Considering the popularity of WhatsApp, I will estimate that around 90% of urban smartphone users and 70% of rural smartphone users have WhatsApp, resulting in around 403 million urban WhatsApp users and 352 million rural WhatsApp users. If this sound reasonable, I would like to further categorize urban and rural WhatsApp users into three age group: under 18, 18-45, and over 45. Should I proceed with this approach?
- Q** Yes, please continue.
- A** Considering the population distribution to be approximately 40% under 18, 40% between 18-45, and 20% over 45, we get:
1. Number of urban WhatsApp user aged <18: 161 million
 2. Number of urban WhatsApp user aged between 18-45: 161 million
 3. Number of urban WhatsApp user aged >45: 81 million
 4. Number of rural WhatsApp user aged <18: 141 million
 5. Number of rural WhatsApp user aged between 18-45: 141 million
 6. Number of rural WhatsApp user aged >45: 70 million
- Does this breakdown seem reasonable?
- Q** Yes, that is a fair segmentation.
- A** Now, I will estimate the average number of chats per day for each segment.
1. Under 18:
Individual chats: 15 per day
Group chats: 5 per day
Total chats: 20 per day
 2. Between 18-45:
Individual chats: 10 per day
Group chats: 3 per day
Total chats: 13 per day
 3. Over 45:
Individual chats: 5 per day
Group chats: 2 per day
Total chats: 7 per day
- Does this seem reasonable?
- Q** Yes.

A Now, calculating the total chats per day for each segment, we get:

1. Chats in age group, under 18: $(161+141) \times 20 = 6.04$ billion
2. Chats in age group, 18-45: $(161+141) \times 13 = 3.93$ billion
3. Chats in age group, over 45: $(81+70) \times 7 = 1.06$ billion

Total Number of WhatsApp chats in a day: $= 6.04 + 3.93 + 1.06 = 11.03$ billion WhatsApp chats per day

Q This was a comprehensive estimate. Thank you.

WHAT WORKED · WHAT WAS MISSING

- ✓ **Asked chats or messages**, and one-to-one or groups: the right first questions.
- ✓ **Stated the approach before starting** and checked it with the interviewer.
- ✓ **Kept individual and group chats separate**, which people use very differently.
- ✗ **“Chat” was never defined.** A conversation you sent a message in? Every message? The whole answer depends on it.
- ✗ **Too many users.** 755 million WhatsApp users is well above the 50–55 crore most estimates give for India in 2026.
- ✗ **40% under 18 is far too high.** WhatsApp’s minimum age in India is 13; teenagers are a much smaller share of users than of the population.
- ✗ **No check.** 11 billion conversations a day means 60–70 billion messages, about two-thirds of everything WhatsApp carries worldwide, from a country with under a fifth of its users.

Guesstimate kit check: 1 Unit ✗ **asked, but never defined** · 2 Approach ✓ · 3 Tree ✓ · 4 Numbers with reasons ✗ · 5 Check ✗ · 6 Range ✗. Skipping card 5 cost the most: WhatsApp’s published 100 billion messages a day would have shown the answer was about four times too high.

CHECK · “40% OF WHATSAPP USERS ARE UNDER 18” DOESN’T HOLD UP

About a quarter of India’s population is under 15, and WhatsApp requires users in India to be at least 13. Many younger teenagers use a parent’s phone rather than their own account. Under-18s are probably closer to 10% of accounts than 40%, and they happen to be the group the answer gives the most chats.

STRONGER ANSWER



Figure 5.21 · The six cards, applied to counting chats. Defining a “chat” (card 1) and anchoring to WhatsApp’s own number (card 5) matter most. Author’s summary.

A 1 · UNIT

I’ll count a “chat” as one conversation, with one person or one group, in which a user sends at least one message that day. Ten messages to your mother count as one chat.

Why this move: “chats” and “messages” differ by about 5–8 times. Defining it out loud stops the interviewer and you from comparing different things.

A 2 · APPROACH

Bottom-up: WhatsApp users in India → those who send something that day → conversations each. Then check against WhatsApp’s published message total.

Why this move: starting from WhatsApp users (not the whole population) removes two uncertain steps, smartphone share and app share, at once.

A 3–4 · TREE AND NUMBERS

About 55 crore WhatsApp users in India. 80% open it on a given day: 44 crore. Of those, 75% send at least one message: 33 crore senders. A sender talks in about 5 one-to-one chats and 3 groups a day: 8 chats. $33 \text{ crore} \times 8 \approx$ **264 crore, about 2.6 billion chats a day** (Figure 5.22).

Why this move: “opens” and “sends” are separate steps because many people only read the family group. Mixing them inflates the count.

A 5 · CHECK

Messages: at about 6 messages per chat, $2.6 \text{ billion} \times 6 \approx 16$ billion messages a day. WhatsApp has said it carries over 100 billion messages a day worldwide, and India is about 18% of its users: $18\% \times 100 \text{ billion} = 18$ billion. Close. The casebook’s 11 billion chats would be 60–70 billion messages, two-thirds of the world’s traffic.

Why this move: a company’s own published number is the strongest anchor you have. Use it every time one exists.

A 6 · RANGE

About **2 to 4 billion chats a day**. The swing factor is chats per sender: at 6 the total is 2 billion, at 12 it is 4 billion. On Diwali or New Year’s Eve it might double for a day.

Why this move: the range comes with its cause, and the festival note shows you know a typical day differs from a peak.

HOW THE ANSWER HANGS TOGETHER

Defining a chat as a conversation fixed the unit. Starting from WhatsApp users, then daily openers, then senders, avoided counting readers as writers. WhatsApp's own 100 billion messages a day, scaled to India's 18% of users, confirmed about 2.6 billion chats. The range names chats per sender as the input to argue about.

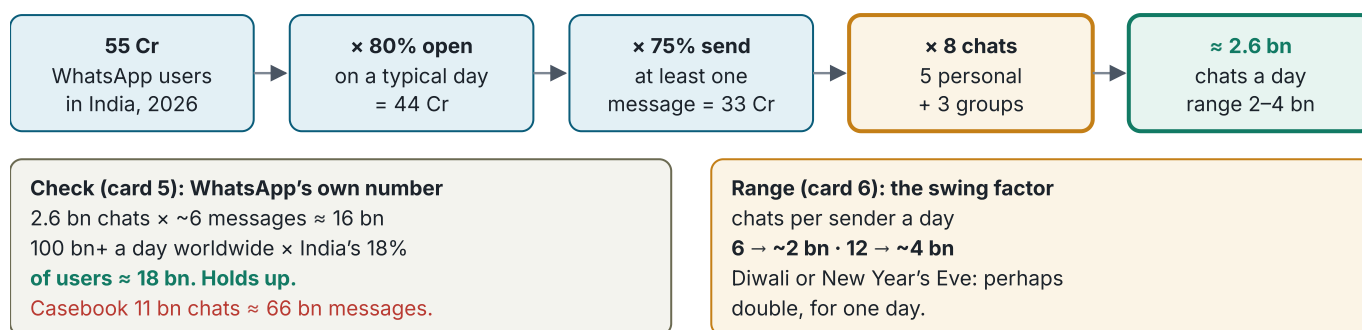


Figure 5.22 · WhatsApp chats a day in India, as an equation tree. The published global message count turns a guess into an estimate you can defend. *India users from 2026 estimates; 100 bn+ messages a day from WhatsApp (last stated 2020); other inputs illustrative.*

THE REUSABLE TRICK

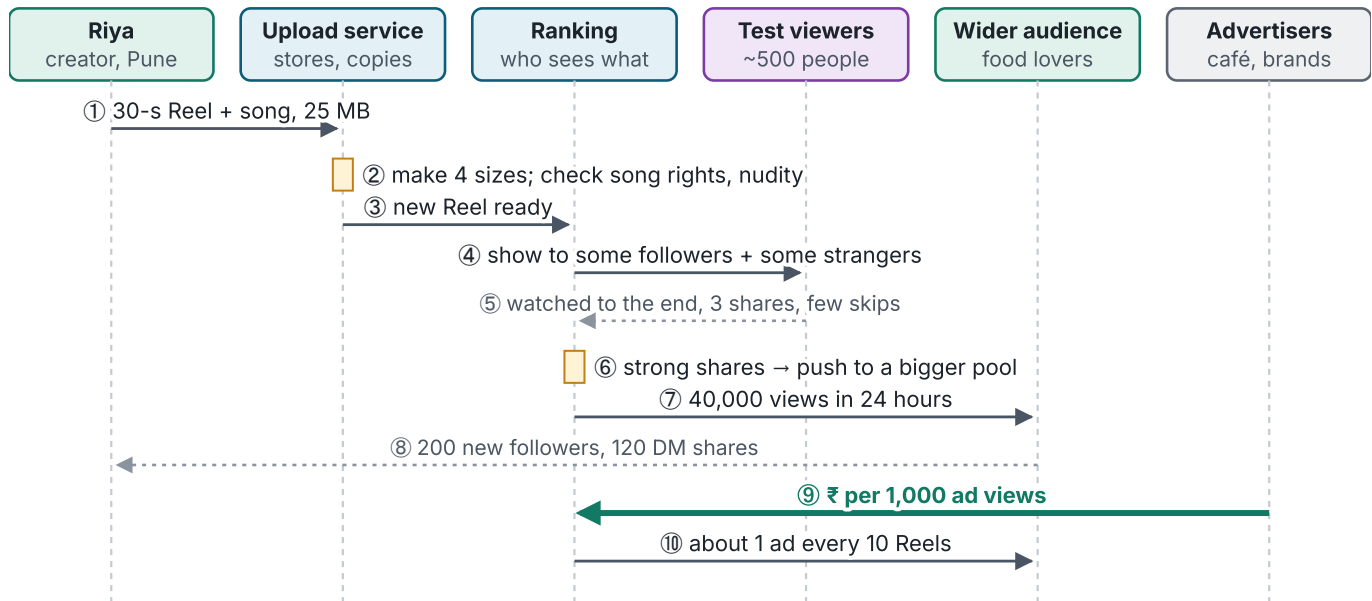
Define the unit so it can't be misread, then check against the company's own number scaled to India's share of users. For messaging, remember the chain: users → openers → senders → conversations → messages. Each arrow can shrink or grow the number several times.

Numbers worth remembering for social and messaging guesstimates

Number	Value (rounded)	Source and date
India's population	140 crore	UN and government estimates, 2025–26
People online in India (regular users)	about 90–95 crore	TRAI subscriptions and industry surveys, 2025–26; unique users are fewer than subscriptions
WhatsApp users in India	about 50–55 crore	Industry estimates, 2026
Instagram users in India	about 40–48 crore	Ad-audience estimates, 2026
YouTube users in India	about 52 crore	2026 estimates
WhatsApp messages a day, worldwide	100 billion+	WhatsApp, last stated 2020
Instagram daily users, worldwide	2 billion	Meta, July 2026
Share of Instagram users who post in a month	a minority; most only watch	90-9-1 pattern; no official figure
Seconds in a day	86,400 (about 1 lakh)	Turns "per day" into "per second": divide by 1 lakh

Instagram: a video feed that keeps friends in the loop

Walkthrough (illustrative persona). Riya, 22, from Pune, films 30 seconds of a vada pav stall at FC Road, adds a trending song in Instagram's Edits app and posts it as a Reel. Within seconds Instagram's servers make four copies at different sizes and check the audio and pictures against its rules. The Reel is first shown to a few hundred people: some of her followers and some strangers who like food videos. They watch it to the end and send it to friends, so Instagram shows it to more. By the next evening it has 40,000 views, 3,000 of them from people who didn't follow her, 120 shares in DMs and 200 new followers (Figure 5.23). Riya earns nothing from Instagram for those views. A café that saw the Reel messages her about a paid post.



Riya earns from the café's paid post, not from Instagram's views.

Figure 5.23 · Behind Riya's Reel. A new Reel is tested on a small audience first (④–⑥); shares count far more than likes in deciding whether it goes wider. Ads slot in between Reels (⑩). Simplified from Instagram's published explanations of Reels ranking (2023–26); numbers illustrative.

Feature	User problem	Why it's designed this way	Metric it moves
Reels	"I'm bored; entertain me"	Full screen, endless, picked by interest rather than by who you follow, so it works from day one	Time spent; daily users
Stories, Close Friends	"I want to share without it staying on my profile"	Gone in 24 hours; a smaller audience lowers the pressure to look perfect	Posts per user
DMs and sharing Reels	"I want to talk about what I watched"	A share button on every Reel sends it into a chat; the friend comes back to watch	Sends per user; returning users
Trial Reels	Creators: "Will my followers hate this?"	Test a Reel on non-followers first, then decide whether to share it with followers	Creators posting weekly
Teen Accounts	Parents: "Is my child safe here?"	Private by default; messages only from people they follow; sleep mode at night	Safety reports; parents' trust; regulatory risk
Product tags, brand deals	Creators: "How do I earn here?"	Tag a product; mark paid partnerships; brands can boost the post as an ad	Creators earning

Core loop: why people come back

Instagram's loop has four steps (Figure 5.24). A **trigger** brings you in, often a friend's message with a Reel. You **act**: open and watch. The **reward** is uneven, and that is the point: one Reel in five is really funny, so you keep swiping for the next one. Then you **invest**: you follow, share or post, which gives Instagram more signals and brings your friends back, starting the loop again for them. The same shape, trigger to action to uneven reward to investment, is Nir Eyal's "Hooked" model, and it also explains why regulators worry about teenagers' screen time.

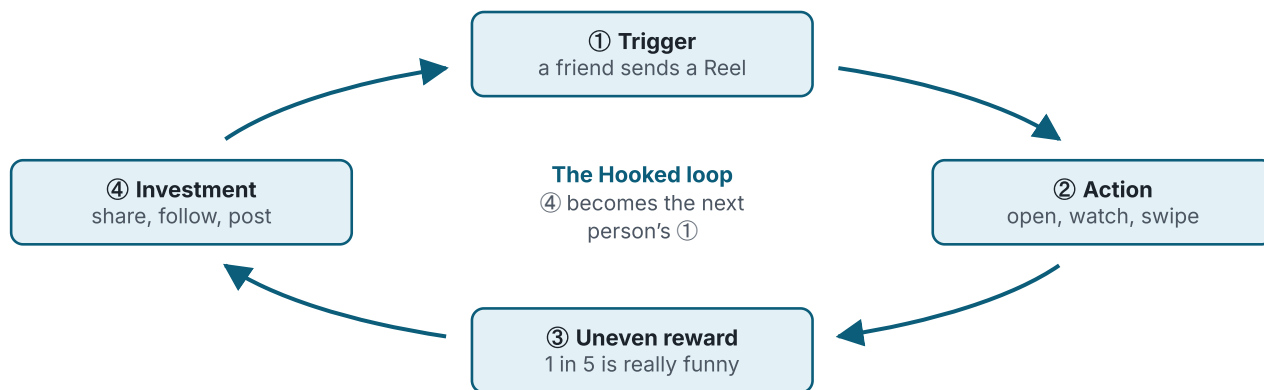


Figure 5.24 · The loop that brings people back. The share in step ④ is also the next person's trigger in step ①, which is why Instagram weights DM shares so heavily. *Author's diagram, after Nir Eyal's Hooked (2014) and Instagram's ranking posts (2025).*

Business model, with numbers

Instagram earns from ads in the feed, Stories and Reels, sold through Meta's auction. Meta doesn't report Instagram's revenue separately, but Instagram is its biggest growth engine: 2 bn daily users (July 2026), and in April–June 2026 Meta's ad impressions rose 14% and the average price per ad 12%. Creators earn mostly from brands, directly. Instagram tries to take a share of that money by letting brands turn a creator's post into an ad.

Moat, in plain words. Your friends and favourite creators are here, and leaving means losing both. Its ranking has learned from billions of swipes, which a new app can't copy quickly. And for Indian creators, Instagram is where brands look first.

NORTH STAR Daily users who share or create, not only watch.	INPUTS Reels watch time · DM shares per user · creators posting weekly · follows from Reels.	GUARDRAILS "Not interested" rate · teen safety reports · ad load · creator churn.
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2025–26 moves. Instagram passed 2 bn daily users. It released Edits, its own video editor, and Trial Reels for creators. Teen Accounts became the default for under-18s. "Your Algorithm" lets users see and change the topics Instagram thinks they like, an answer to "my feed feels random".

STRENGTHS

- ✓ India's default app for creators and brands
- ✓ Friends and entertainment in one place
- ✓ Meta's ad machine behind it

WEAKNESSES

- ✗ Feels less personal as strangers' Reels take over
- ✗ Low revenue per Indian user
- ✗ Teen safety and screen-time criticism

REAL-LIFE EXAMPLE · THREADS: BORROWING A NETWORK TO SKIP THE COLD START

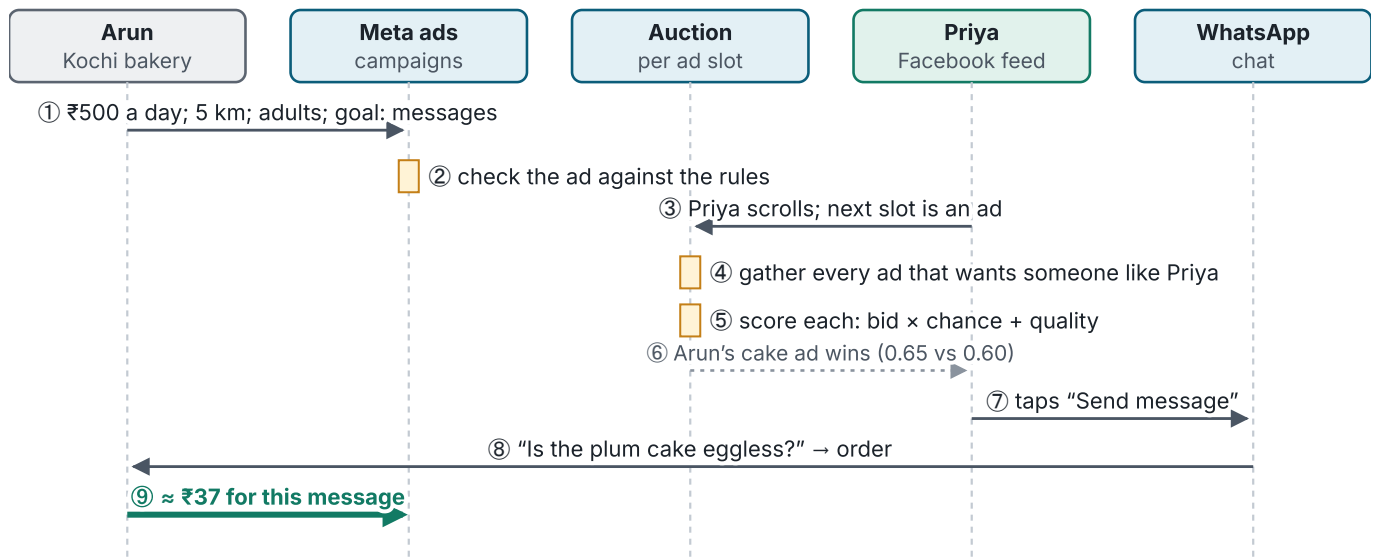
Meta launched Threads, a text app like X, in July 2023. New users could sign in with Instagram and follow everyone they already followed there in one tap, so nobody opened an empty app. 10 crore people signed up in five days. Many then drifted away: a borrowed network gets sign-ups, but only posts worth reading bring people back. Meta added messages, communities and a feed users can steer, and Threads reached **500 mn monthly users in June 2026**. The lesson for any new social product: bring the network, then give it a reason to stay.

What I'd build next

1. Indian creators sell through DMs and payment screenshots → a creator shop inside DMs with UPI payment and order tracking → creators earning each month; complaints per 1,000 orders.
2. People want to know what's happening in their own city this weekend → a "Near me" Reels feed of events, food and local businesses, with a map → local sessions per user; ad spend from local businesses.
3. Parents can't see how their teenager uses the app → a weekly summary for parents (time spent, new follows), agreed by the teen → share of teen accounts with a parent linked; teen retention.

Facebook and Meta's ad machine

Walkthrough (illustrative persona). Arun runs a bakery in Kochi. Two weeks before Christmas he opens Meta's ad tool and spends ₹500 a day to show a photo of his plum cakes to adults within 5 km, with a "Send message" button. That evening Priya, 34, scrolls Facebook. In the few milliseconds before the next post appears, Meta holds an auction for that one slot among every advertiser who wants someone like Priya. Arun's ad wins. Priya taps "Send message", a WhatsApp chat with the bakery opens, and she orders a 1 kg cake. Meta charges Arun about ₹37 for that message (Figure 5.25).



Arun pays for the action he chose (a message), not for the view.

Figure 5.25 · Behind one ad on Priya's feed. The auction (④–⑥) runs for every ad slot, billions of times a day. Arun pays only when Priya does what he paid for, here sending a message. *Simplified from Meta's published description of its ad auction; amounts illustrative.*

How the auction picks a winner

Meta doesn't simply show the highest bid. It ranks ads by **total value** = the advertiser's bid for an action × the chance that this person does it + a quality score (which falls if people hide or report the ad). Arun bids ₹40 for a message, and Meta predicts a 1.5% chance Priya sends one: ₹40 × 1.5% = ₹0.60, plus a small quality bonus, 0.65. A phone brand bids ₹8 per click at 5%: 0.40, plus 0.10 = 0.50. A loan app bids ₹100 per install at 0.8%: 0.80, but its ads are often reported, so its quality score is -0.20: 0.60 (Figure 5.26). Arun wins with 0.65, and pays only enough to beat the next ad, 0.60, which works out to about ₹37 per message instead of his ₹40 bid.

Advertiser	Bid	Chance Priya does it	Bid × chance	Quality	Total value
Arun's bakery	₹40 a message	1.5%	0.60	+0.05	0.65 wins
Phone brand	₹8 a click	5%	0.40	+0.10	0.50
Loan app	₹100 an install	0.8%	0.80	-0.20	0.60

Arun pays just enough to beat 0.60: $(0.60 - 0.05) \div 1.5\% \approx ₹37$ per message, not his full ₹40 bid.

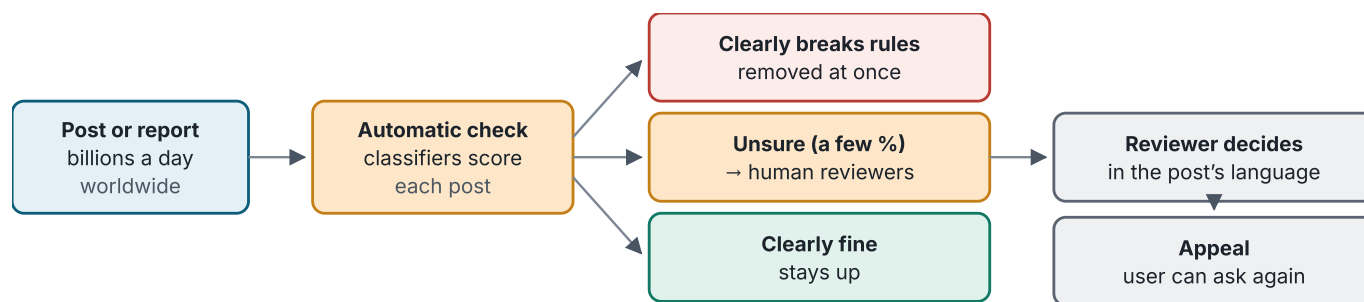
The loan app bids the most but loses: Priya is unlikely to install, and people report its ads.

Figure 5.26 · One ad slot, three bidders. The highest bid (the loan app's ₹100) loses because Priya is unlikely to install it and people report its ads. Relevance and quality beat money alone. *Formula from Meta's published auction description (total value = bid × estimated action rate + ad quality); numbers illustrative.*

Why design it this way? If the biggest bidder always won, feeds would fill with ads people hate, they would scroll less, and every advertiser would lose. Weighting by the chance of action makes a small bakery competitive against a national brand for the people nearby who might actually want cake. It is also why Meta's better AI models raise its price per ad: when ads convert better, advertisers bid more.

Keeping the feed safe: the moderation pipeline

Billions of posts a day can't all be checked by people. Automatic classifiers score each post; posts that clearly break the rules are removed at once, clear cases pass, and uncertain ones go to human reviewers, who work in many Indian languages (Figure 5.27). India's IT Rules 2021 make large platforms publish monthly compliance reports and act on government and court orders within fixed times. The sum that sets the size of the review team: if 2 lakh posts a day in India need a human look, and a reviewer handles about 400 a day, Meta needs $2,00,000 \div 400 = 500$ reviewers on duty, about 700 after shifts and leave (illustrative).



Team size (illustrative): 2 lakh posts a day need a human look in India ÷ 400 per reviewer a day = 500 on duty, about 700 with shifts and leave. Tighten a rule, and posts move from “fine” to “unsure”: the team must grow.

Figure 5.27 · How a reported post is handled. Machines take the clear cases; people take the grey ones. Every rule change moves posts between the boxes, and so changes the size of the review team. *Simplified from Meta's Transparency Center descriptions (2024–26); volumes illustrative.*

Feature	User problem	Why it's designed this way	Metric it moves
Groups	“People who share my interest or problem”	Admins, rules, topics; where much of Facebook's daily use happens	Daily active group members
Marketplace	“Buy and sell used things nearby”	Local listings and chat with the seller; high intent, so good for ads	Listings that sell
Friends tab, video feed	“I only want my friends” / “entertain me”	A friends-only tab beside an AI-ranked video feed: both jobs in one app	Time spent; young adults' share
Click-to-WhatsApp ads	Small shops: “My customers live on WhatsApp”	The ad opens a chat instead of a website	Messages started; advertiser spend

Business model, with numbers

Meta's revenue in April–June 2026 was **\$60.8 bn**, up 28%, almost all from ads across Facebook, Instagram, Messenger and WhatsApp. 3.60 bn people used at least one Meta app each day. Its automated campaign tool, Advantage+, runs at about \$75 bn a year. Meta also spends heavily: Reality Labs (glasses and headsets) lost \$4.6 bn that quarter, and it plans \$130–145 bn of spending on data centres for AI in 2026.

Moat, in plain words. Three billion people and the data on what they respond to, plus an auction that small businesses can use for ₹500 a day. Advertisers go where the customers and the results are. The risk is age: Facebook's users are getting older, which is why the young go to Instagram.

NORTH STAR Daily users who interact with friends, groups or Marketplace.	INPUTS Group posts · Marketplace messages · video time · ad conversion rate.	GUARDRAILS Scam reports per 1,000 Marketplace chats · hide and report rate on ads · share of daily users under 30.
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2025–26 moves. A Friends tab and a feed rebuilt around recommended video. Messenger was folded back into Facebook: messenger.com shut in April 2026. In the US, Community Notes replaced third-party fact-checkers in 2025.

STRENGTHS

- ✓ The largest audience in the world
- ✓ An ad system small shops can use
- ✓ Groups and Marketplace give daily reasons to visit

WEAKNESSES

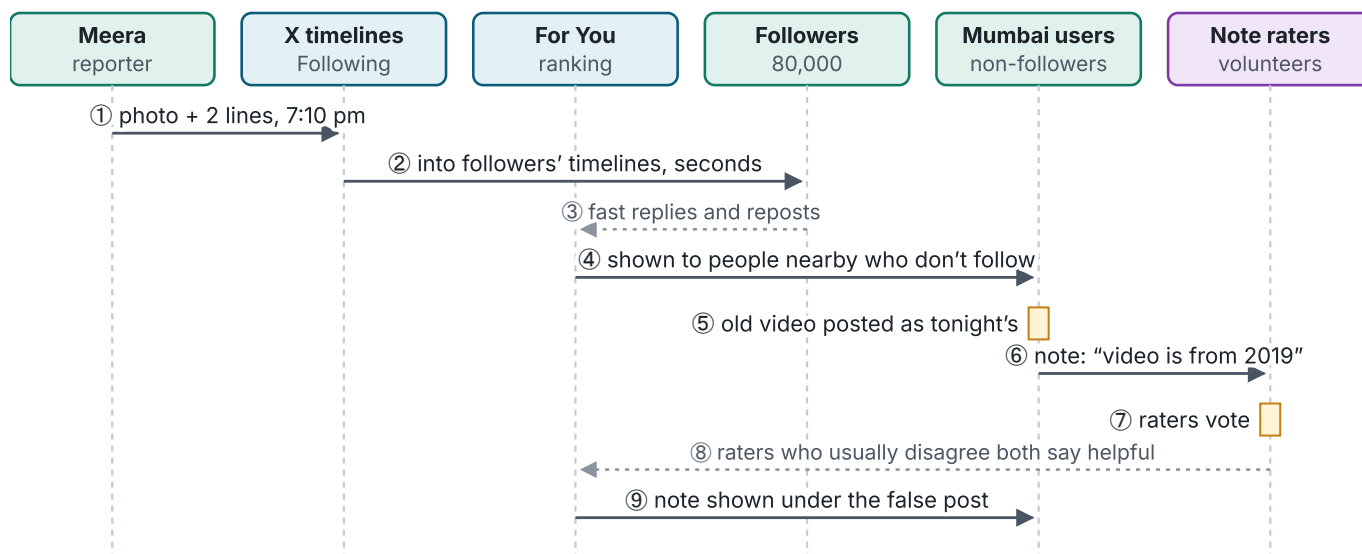
- ✗ Ageing users; weak with under-25s
- ✗ Scams on Marketplace and in ads
- ✗ Constant rulings on content and data, in India and abroad

What I'd build next

1. Buyers fear being cheated on used goods → verified sellers plus UPI payment held until the buyer confirms pickup → listings that sell; scam reports per 1,000 deals.
2. Small shops don't know if their ₹500 a day worked → a weekly “what your ads brought in” message on WhatsApp, with orders from chats counted → small advertisers still spending after 90 days.

X: the fastest place for news, and a shrinking business

Walkthrough (illustrative persona). Meera, a reporter in Mumbai, sees smoke over a railway yard at 7:10 pm and posts a photo with two lines. Her 80,000 followers get it in their “Following” timeline within seconds. X's “For You” ranking notices fast replies and reposts, and shows it to people in Mumbai who don't follow her. Within an hour, another account posts an old video from a different fire, claiming it is tonight's. Readers who add context write a note: “This video is from 2019.” The note appears under the post only after raters who usually disagree with each other both mark it helpful (Figure 5.28).



Speed is X's strength; a note arriving an hour late is its weakness.

Figure 5.28 · Behind a breaking-news post, and the note on a false one. Followers get the post directly (2); strangers get it only if the ranking picks it (3–4). A note is shown only when people from different sides agree (7–9). *Simplified from X's published Community Notes and ranking descriptions (2023–26); numbers illustrative.*

The clever part: notes that both sides must agree on

Most fact-check systems ask a few experts, and half the audience distrusts them. X's **Community Notes** ask users instead, but a note isn't shown just because many people like it; that would let the bigger side win every time. The system learns which raters usually disagree, and shows a note only when people from *both* groups rate it helpful (Figure 5.29). Designers call this **bridging**. Meta copied the idea for Facebook, Instagram and Threads in the US in 2025.

Each dot is a rater who marked the note helpful

raters who usually lean one way

raters who usually lean the other way

Note A: 9 helpful votes



Hidden: all votes from one side
popular with one group only

Note B: 5 helpful votes



Shown: helpful to both sides
people who usually disagree agree



Figure 5.29 · Why popularity alone doesn't publish a note. Note A has more helpful votes, all from one side, and stays hidden. Note B has fewer votes, but from both sides, and is shown. *Author's diagram of X's open-source Community Notes ranking; votes illustrative.*

Business model, with numbers

X earns from ads and from subscriptions (Premium, for a blue tick and more reach). After Elon Musk bought Twitter in October 2022 for \$44 bn and cut about 80% of its staff, many big advertisers left. Ad revenue fell from \$4.4 bn in 2022 to about \$2 bn in 2025. X now sits inside the SpaceX group through its merger with xAI, and a 2026 filing showed **550 mn monthly users**, fewer than the 600 mn X had claimed, and about 6.3 mn paying subscribers. It is also testing X Money, for payments.

Moat, in plain words. Journalists, politicians, investors and sports fans still post here first, so news breaks here first. That habit is hard to copy, but Threads and Bluesky are trying. In India, X is where governments and ministers announce things, which also makes it a frequent subject of takedown orders: X challenged India's Sahyog portal for such orders in the Karnataka High Court and lost in September 2025.

NORTH STAR Daily users who read or post during live events.	INPUTS Posts and replies per user · notes shown on viral posts · Premium sign-ups.	GUARDRAILS Advertiser brand-safety complaints · spam and bot share · time from false post to note.
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STRENGTHS

- ✓ Real-time news and public debate
- ✓ Community Notes, a design others copy
- ✓ Grok AI built in

WEAKNESSES

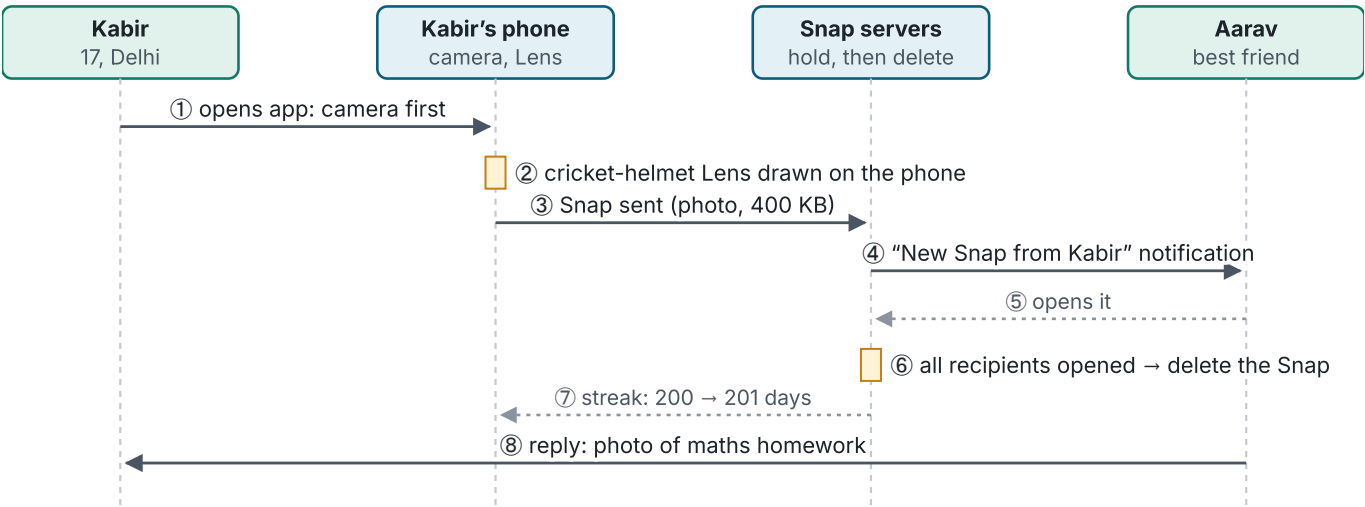
- ✗ Ad revenue half of 2022's
- ✗ Decisions depend heavily on one owner
- ✗ Bots, spam and brand-safety worries

What I'd build next

- 1. During a crisis, people can't tell rumours from reports → a “verified sources” view that shows only news organisations and posts with notes → daily users during breaking news; time from false post to note.
- 2. Indian cricket fans scatter across apps during matches → official match rooms with live scores and highlights in Hindi, Tamil and Bengali → match-day daily users; ad revenue per match.

Snapchat: the camera company

Walkthrough (illustrative persona). Kabir, 17, in Delhi, has a 200-day “streak” with his best friend Aarav: they have sent each other a Snap every single day. At 10:40 pm he remembers. The app opens straight to the camera, not a feed. He picks a cricket-helmet Lens, which the phone itself draws onto his face in real time, snaps a photo and sends it to Aarav. Snap’s servers hold the photo until Aarav opens it, then delete it. The streak counter moves to 201. Aarav replies with a picture of his maths homework (Figure 5.30).



Unopened Snaps are deleted after 30 days; opened ones are gone from Snap’s servers.

Figure 5.30 · Behind one Snap. The Lens runs on the phone (②), so it costs Snap no server time. The servers keep a Snap only until it is opened (④–⑥). The streak (⑦) is a daily reason to open the app. *Simplified from Snap’s published privacy and product pages; illustrative.*

Feature	User problem	Why it’s designed this way	Metric it moves
Opens to the camera	“I want to share this moment now”	Creating is the default, not scrolling; one big button: tap for photo, hold for video	Snaps sent per user
Snaps that disappear	“I don’t want every silly photo kept forever”	Deleted after viewing; a screenshot alerts the sender	Private sends; trust among teens
Lenses	“A plain photo is boring”	AR effects make ordinary moments worth sending; brands sponsor Lenses	Lens plays; ad revenue
Streaks	“Stay close to my best friends”	A visible daily count; losing it hurts	Daily users; days active per month
Snapchat+	Fans: “I want extras”	A paid tier with early features and custom icons	Subscribers; “other revenue”

Business model, with numbers

In April–June 2026 Snap earned about **\$1.6 bn**, up 19%. Most of it came from ads: Lens ads, Stories ads and ads in Spotlight, its short-video feed. Its “other revenue”, mostly Snapchat+ subscriptions, grew 85% to **\$316 mn** (Figure 5.31). Snap has about 1 bn monthly users and added 10 mn daily users that quarter, most of them outside North America and Europe, with India a big contributor. That is Snap’s puzzle: its growth is where each user earns it least.

Snap, one quarter

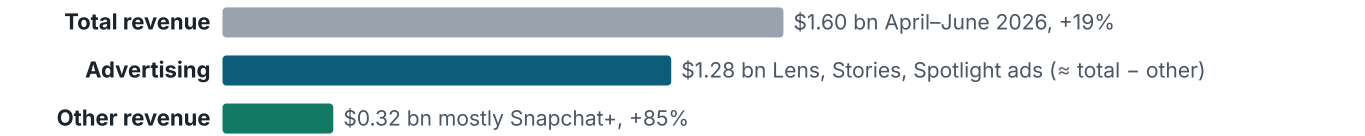


Figure 5.31 · Where Snap’s money came from, one quarter. Subscriptions are small but growing fastest, a hedge against low ad prices in markets like India. *Snap Q2 2026 results (advertising estimated as total minus other revenue, rounded).*

Moat, in plain words. Your closest friends and your streaks are here. Snap is private and playful, not about follower counts, which is why teenagers use it to talk and Instagram to show off. It also has one of the best AR engines, with

tools that let outside creators build Lenses.

NORTH STAR Daily users who send at least one Snap.	INPUTS Snaps per user · Lens plays · streaks active · Snapchat+ sign-ups.	GUARDRAILS Teen safety reports · revenue per user by region · stress about streaks (surveys).
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STRENGTHS

- ✓ Close friends, daily habit
- ✓ Leader in AR Lenses
- ✓ Subscriptions growing fast

WEAKNESSES

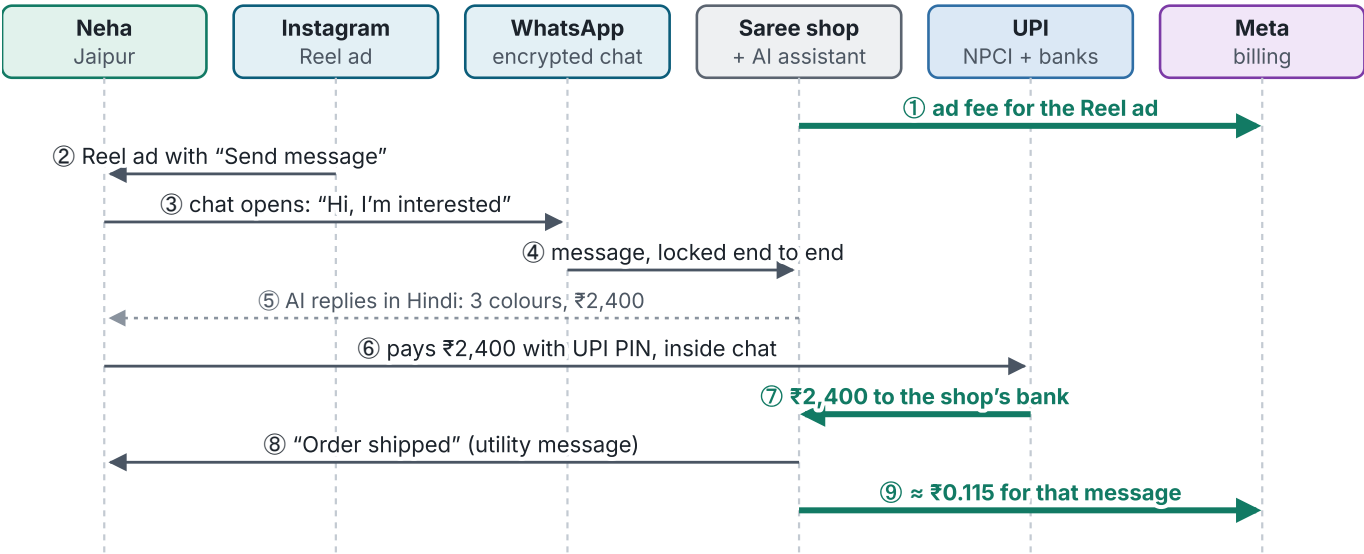
- ✗ Low ad revenue per user where it grows
- ✗ Streaks can feel like pressure
- ✗ Instagram copies its best ideas (Stories, 2016)

What I'd build next

- Indian users are active but earn Snap little → self-serve sponsored Lenses for Diwali, IPL and college fests, priced for local brands at ₹5,000 → India revenue per daily user; Lens plays per user.
- A lost streak makes teenagers anxious → one free “streak freeze” a month → daily users (guardrail: stress reported in surveys).

WhatsApp: India’s default way to talk, and now to buy

Walkthrough (illustrative persona). Neha, 31, in Jaipur, is watching Reels when an ad from a Sanganeri saree shop shows a “Send message” button. She taps it and WhatsApp opens a chat with the shop, with “Hi, I’m interested” already typed. The shop’s AI assistant replies in Hindi within seconds: three colours, ₹2,400, delivery in four days. She picks one from the catalogue in the chat, pays by UPI without leaving WhatsApp, and gets an “order shipped” message two days later. Meta earned twice: once for the ad, and a few paise for the shipping update (Figure 5.32).



Meta earns twice from the shop, never from Neha; it can’t read the chat itself.

Figure 5.32 · Behind Neha’s saree order. The chat itself is encrypted end to end, so WhatsApp can’t read it (③–⑤). Meta is paid by the shop, not by Neha: for the ad (①) and per business message (⑨). Simplified from Meta’s WhatsApp Business documentation; India prices approximate (January 2026); amounts illustrative.

Feature	User problem	Why it’s designed this way	Metric it moves
End-to-end encryption	“Is anyone reading my chats?”	Only the people in the chat hold the keys; on by default since 2016	Trust; long-term retention
Voice notes	“Typing Hindi on a phone is slow”	Hold to talk; now with transcripts	Messages per user
Groups and Communities	“Our housing society needs one place for notices”	Groups up to 1,024, grouped under a Community with an announcements channel	Active groups
Status and Channels	“I want to share with everyone, or follow a creator”	Broadcast features kept in a separate Updates tab, away from chats	Updates-tab users; ad revenue
Business app, AI agents	Shops: “I can’t answer 500 chats a day”	Catalogue, quick replies, AI that answers and hands over to a person	Paying businesses
UPI payments	“Pay without leaving the chat”	Built on UPI; NPCI removed its user cap on WhatsApp Pay in December 2024	Payments started in chats

How WhatsApp earns without ads in chats

Users pay nothing. The money comes from three places. **Business messages**, priced per message by type (Figure 5.33). **Click-to-WhatsApp ads** on Facebook and Instagram, which open a chat instead of a website. And since 2025, **ads and paid subscriptions in the Updates tab**, where Status and Channels live, never in personal chats. Meta’s “other revenue”, mostly business messaging, reached **\$1 bn** in April–June 2026, up 73%, and more than 10 lakh businesses use its AI agents each week.

Price per message to a user in India, from January 2026 (approx.)

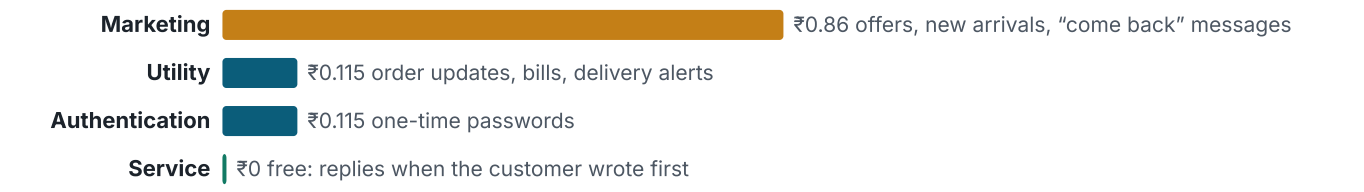


Figure 5.33 · What a business pays WhatsApp per message in India. Marketing is priced highest to keep offers rare; replies to a customer who wrote first are free, which rewards good service. *Meta WhatsApp Business pricing, India, billed in rupees from January 2026 (approximate).*

Moat, in plain words. Everyone you know is on it, reachable by the phone number already in your contacts, and years of chats and photos live in it. That makes it one of the strongest positions in consumer technology. The risks are misinformation spreading through forwards, pressure from governments to trace messages, and the temptation to earn so much from businesses that personal chats fill up with offers.

NORTH STAR Daily users who send a message or make a call.	INPUTS Messages and calls per user · active groups · business conversations per week · Updates-tab reach.	GUARDRAILS Spam and business-block reports · share of users muting business chats · forward-limit breaches.
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2025–26 moves. Per-message pricing (July 2025) and rupee billing (January 2026). Ads and paid Channel subscriptions in the Updates tab. Business AI agents at scale. In June 2026 Kunal Shah, founder of CRED, was named head of WhatsApp, a sign that payments and commerce will get more attention.

REAL-LIFE EXAMPLE · TELEGRAM, THE PUBLIC MESSENGER

Telegram made the opposite choices, and each follows from one decision: it is built for people who *don’t* know each other. It has over 1 bn monthly users, about 15 mn paying Premium subscribers, and India was its biggest download market in 2025. It made its first annual profit, \$547 mn, in 2024, and \$870 mn of revenue in January–June 2025.

	WhatsApp	Telegram
Built for	People who know each other	People who don’t: channels, big communities
Encryption	End to end for every chat, by default	Stored on its servers by default; end to end only in “secret chats”
Group size	Up to 1,024; Channels for broadcast	Groups up to 2 lakh; channels with no limit
How you’re found	Your phone number	A username or public link
Earns from	Businesses, Updates-tab ads	Premium, channel ads, crypto (TON)
Biggest risk	Over-earning from users’ trust	Piracy, scams, weak moderation (its founder was arrested in France in 2024)

STRENGTHS	WEAKNESSES
✓ Nearly every smartphone user in India	✗ Little revenue per user
✓ Private by default, and trusted for it	✗ Forwards spread rumours it can’t see
✓ Businesses already talk to customers here	✗ Weaker than Telegram for large public groups

What I’d build next

- 1. Offers from businesses crowd out personal chats → a “Businesses” folder with one-tap unsubscribe and a monthly cap per brand → business blocks per 1,000 messages (down); order-update read rate (guardrail: marketing revenue).
- 2. Kirana owners take orders in chat and lose track → turn any customer message into an order card with items, total and a UPI request → weekly active small businesses; payments started from chats.
- 3. Housing societies chase maintenance fees in groups → fee requests with UPI and a paid/unpaid list for the committee inside Communities → active society Communities; share of fees paid on time.

Side by side

	Instagram	Facebook	X	Snapchat	WhatsApp
Main job	Entertain me; share with friends	Groups, Marketplace, friends’ news	What’s happening now; argue about it	Talk to close friends in pictures	Talk to people I know; buy from shops

Whose posts you see	Mostly strangers', picked by interest	Friends, groups, recommended video	Followed accounts plus "For You"	Close friends; Spotlight for strangers	Only people and groups you chose
Earns from	Ads	Ads	Ads, subscriptions	Ads, Snapchat+	Business messages, click-to-chat ads, Updates tab
Latest number	2 bn daily users (Jul 2026)	3 bn+ monthly users	550 mn monthly users	~1 bn monthly users	India 50 crore+ users
Moat	Friends and creators; ranking	Scale; ad auction	News habit of public figures	Close friends; streaks; AR	Everyone's phone book; trust
Biggest threat	YouTube Shorts; teen-safety rules	Ageing users	Advertisers leaving	Low revenue per user	Losing new groups; traceability demands

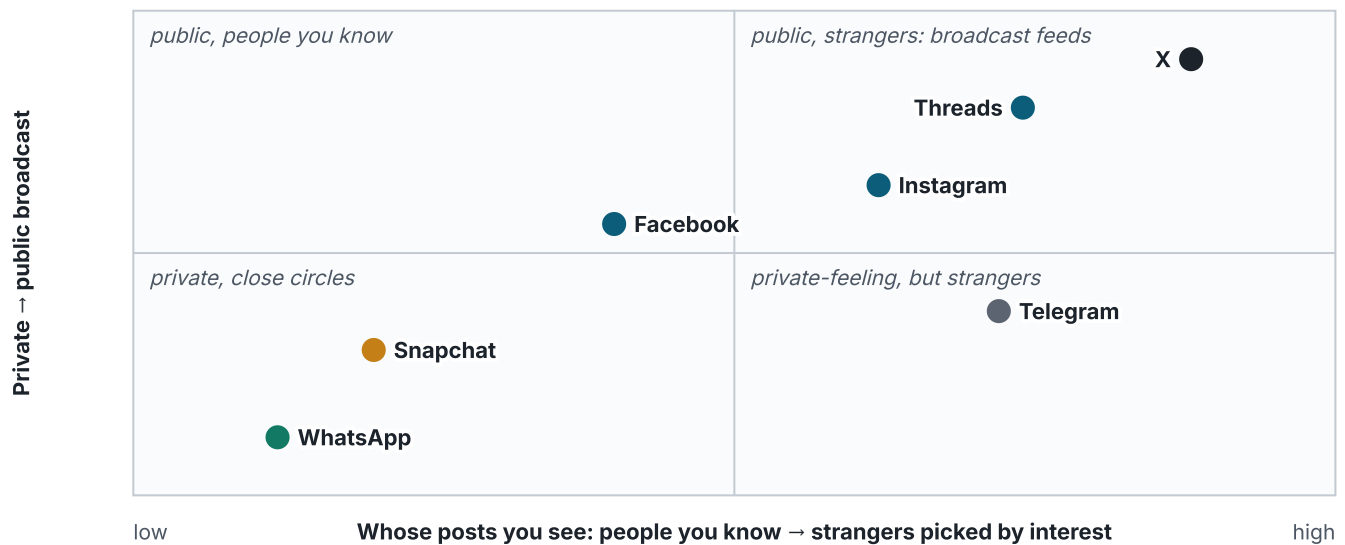


Figure 5.34 · Who you hear from, and how public it is. Messaging apps sit bottom left: private, with people you know. Feed apps have moved up and right, towards strangers and broadcast, because that is where attention (and ad money) is. Telegram sits in between: private-feeling, but built for strangers. *Author's placement; positions are judgement, not measured values.*

IN THE INTERVIEW · HOW THESE PRODUCTS COME UP

- **“How does Meta make money from WhatsApp?”** Business messages by type, click-to-WhatsApp ads, and the Updates tab. Never ads in chats.
- **“Why did Instagram beat Snapchat?”** It copied Stories in 2016 and gave them to a much bigger network; the features were similar, the audience wasn't.
- **“Why do small shops love Meta ads?”** ₹500 a day, targeting by distance, and a chat instead of a website.
- **“Is Community Notes a good design?”** Yes: it needs agreement across sides, so neither side can outvote the other. Its weakness is speed: a note often arrives after a false post has spread.

System design IV: one WhatsApp message, end to end

9:14 pm. Farhan's aunt records a voice note, "Beta, the wedding is on the 14th", and sends it to the 47-member family group. Farhan's phone is in his pocket with WhatsApp closed. His cousin Zoya is on a flight to Dubai with her phone in flight mode. Within a second his aunt sees one grey tick, then two. Farhan's phone buzzes. Three hours later, when Zoya lands, the note is waiting for her. None of this feels special, and that is the achievement: it happens more than 100 billion times a day. This section follows that one message, one step at a time.

One post office, ten ideas

A messaging app works much like a well-run post office that delivers registered letters. Keep this table in mind; each idea in this section is one row of it.

Idea	The post office	WhatsApp
Always-on connection	A postman who walks your lane all day, instead of you walking to the post office every ten minutes to ask "anything for me?"	Each phone keeps one open line to WhatsApp's servers; messages are pushed down it
Receipts and ticks	Registered post: a receipt at the counter, an acknowledgement card when delivered	✓ the server has it; ✓✓ your phone has it; blue ✓✓ you opened it
Store and forward	Letters held at the post office while you are away, returned after a month	Messages wait on the server until your phone is back, up to 30 days
Order and duplicates	Numbered pages; if page 3 comes twice, you throw one away	Every message has an ID and a place in the chat; repeats are ignored
Group fan-out	A wedding card printed once, one copy posted to each relative	The server copies one message into 46 members' inboxes
Push notifications	The society guard rings your intercom: "parcel at the gate"	Apple's and Google's services wake the sleeping app
Media by reference	A big parcel kept at the post office; you get a collection slip	A video is uploaded once; the message carries a link and a key
Presence	The light in your window that says you're home	"Online", "typing...", "last seen": sent live, never stored
Forward limits	A chain letter stamped "copied many times", accepted for one address at a time	A forward counter travels with the message; after five hops, one chat at a time
When a server dies	The lane's postman falls ill; another takes over and the sorting room is untouched	Phones reconnect to another server, spread out over a minute; queued messages survive

The load, in numbers

100 billion messages a day $\div$ 86,400 seconds $\approx$ **11.6 lakh messages every second**, on average. At midnight on New Year's Eve it can be several times that. India is about 18% of WhatsApp's users, so roughly 2 lakh messages a second on a normal evening. Every message also creates two or three smaller ones (the receipts behind the ticks). And at 9 pm perhaps 25 crore Indian phones are connected at once. If one server can hold about 20 lakh open connections, that is 25 crore $\div$ 20 lakh = 125 servers just to keep the lines open, about 250 with spare room (illustrative). Text is cheap to carry; the hard parts are the number of open connections, the bursts, and the photos and videos.

1. An always-on connection, not constant asking

Everyday picture. Waiting for an important letter, you could walk to the post office every ten minutes and ask. Most trips would be wasted, and you would be exhausted by evening. Or the postman who walks your lane could simply hand it to you when it arrives.

How it works. When WhatsApp is open or running in the background, the phone opens **one long-lived connection** to a chat server and keeps it open. The server pushes messages down it the moment they arrive. Mobile networks close lines that stay silent too long, so the phone sends a tiny **heartbeat** (a keep-alive message of a few bytes) every few minutes. The other way, **polling**, means the phone asks "anything new?" again and again (Figure 5.35). Polling every 10 seconds is $86,400 \div 10 = 8,640$ requests a day per phone, almost all answered "nothing", and each one wakes the phone's radio and drains the battery. A heartbeat every 4 minutes is $1,440 \div 4 = 360$ a day: 24 times fewer, and a message arrives in under a second instead of waiting up to 10.

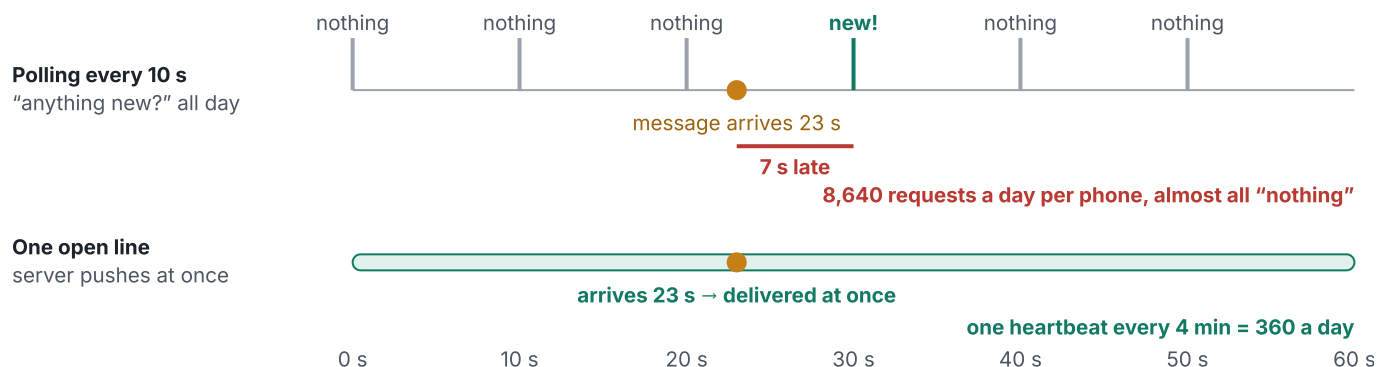


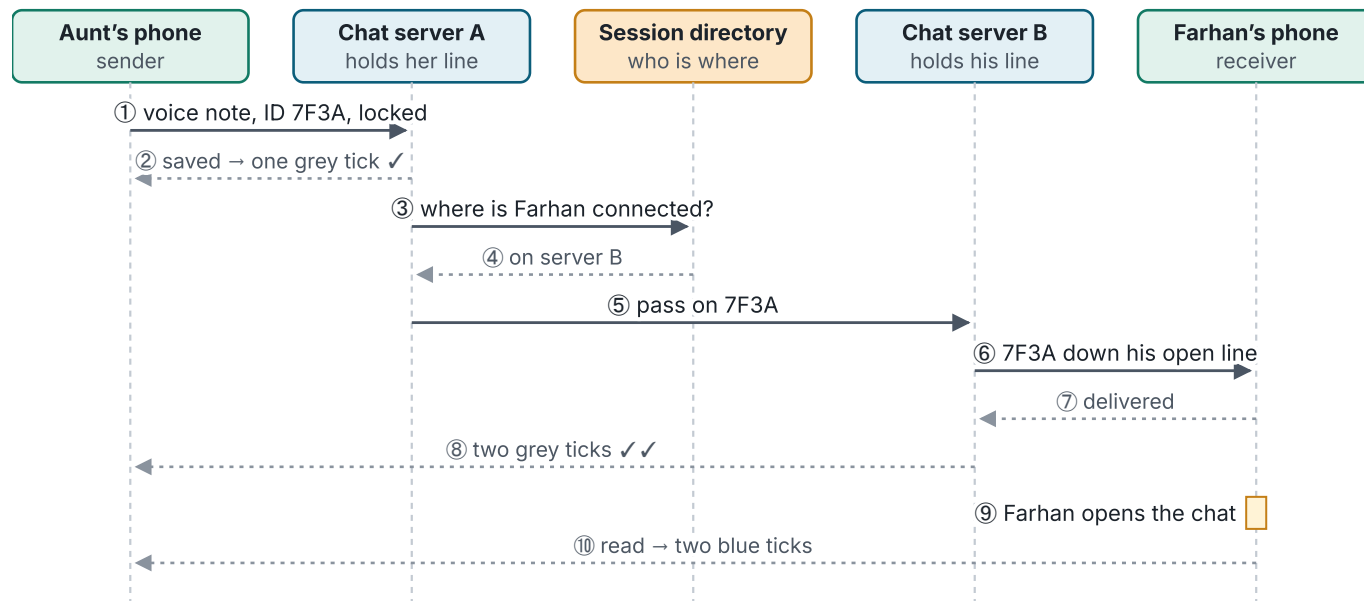
Figure 5.35 · Asking all day, or keeping one line open. Polling wastes thousands of requests and still delivers late; an open connection delivers at once and only needs a small heartbeat to stay alive. *Illustrative intervals; the pattern is how chat apps generally work.*

Real example. WhatsApp's engineers wrote in 2012 that one of their servers was holding more than 20 lakh open connections at the same time. They used Erlang, a programming language first built by Ericsson for telephone exchanges, which were designed for exactly this: millions of calls waiting, few of them active. **PM decision:** the heartbeat interval trades battery against speed of noticing a dropped connection. It also decides how quickly "online" changes to "last seen". On cheap phones with small batteries, a slightly longer interval can be the difference between a loved app and an uninstalled one.

2. The journey of one message, and the three ticks

Everyday picture. Registered post. The counter clerk gives you a receipt when she accepts the letter. An acknowledgement card comes back when it is delivered. If the person phones to say they have read it, you know that too.

How it works (Figure 5.36). ① The aunt's phone gives the voice note a unique ID, locks it with encryption and sends it on its open connection. ② The chat server saves it and replies "got it": **one grey tick**. ③ The server looks up which server holds Farhan's connection, in a **session directory** (a constantly updated list of which phone is connected where). ④ It passes the message there and down Farhan's line. ⑤ Farhan's phone replies "delivered", which travels back: **two grey ticks**. ⑥ When Farhan opens the chat, his phone sends "read": **two blue ticks**. Each receipt is itself a tiny message, so one message sent to one person creates about three messages' worth of work.



Every receipt is a tiny message of its own: one message sent creates about three.

Figure 5.36 · What each tick means. One tick is a promise from the server; two ticks, from the other phone; blue, from the other person. Every receipt is a small message travelling back. *Simplified from how WhatsApp describes message delivery and receipts; server names illustrative.*

Real example. Blue ticks arrived in November 2014, and many users hated being seen to have "read and not replied". Within weeks WhatsApp let people switch read receipts off, with a fair rule: if you hide yours, you can't see anyone else's. Groups always show who read a message, under "Message info". **PM decision:** each tick is a promise, so it must be exactly true. Showing two ticks before the other phone confirms would be faster, and it would be a lie that users would discover. Privacy settings that cut both ways, like read receipts, feel fair and avoid endless arguments.

3. Store and forward: when the other phone is off

Everyday picture. You go to your village for a week. The post office holds your letters and hands them over when you are back. If you never come, they go back to the sender after a month.

How it works. If Zoya's phone isn't connected, the server puts the message in her **offline queue**, stored on disk, with copies on more than one machine so a single failure can't lose it. Her aunt sees one tick. When Zoya lands and her phone reconnects, the server sends everything in her queue in order, waits for each "delivered", then deletes it. WhatsApp keeps delivered messages on its servers for no longer than it takes to deliver them, and deletes undelivered ones after 30 days (Figure 5.37).

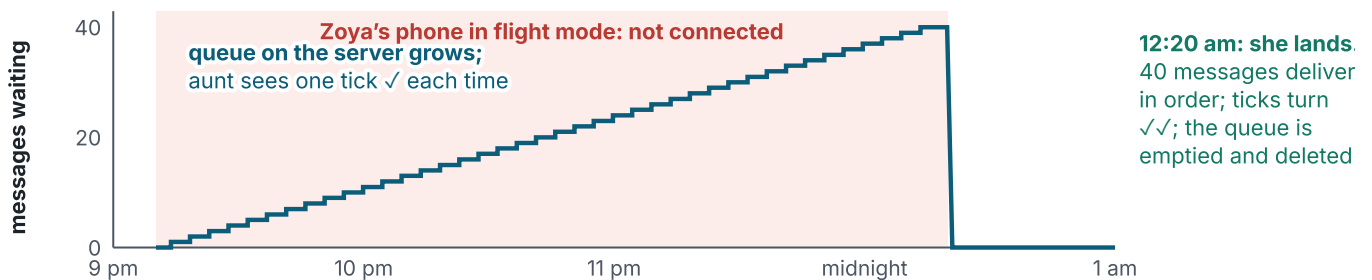


Figure 5.37 · Zoya's three hours in the air. Messages pile up in her queue on the server; the aunt sees one tick for each. When Zoya's phone reconnects, the queue empties in order, and the ticks turn double. *Illustrative timeline; 30-day limit from WhatsApp's help centre.*

Order and duplicates. Phones lose signal in lifts and trains, then try again, so the same message can be sent twice. Because every message has an ID, the receiving phone keeps a list of IDs it has seen and quietly drops repeats, like throwing away a second copy of page 3. The server numbers messages within each chat, so they appear in the order they were sent. The promise users feel is simple: everything arrives once, in order, even if the network was a mess.

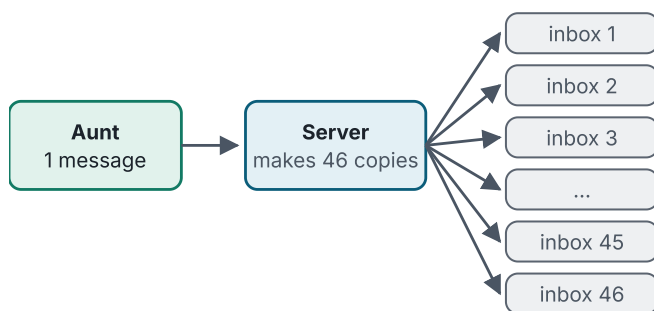
Real example. The clock icon next to a message sent in a metro tunnel turns into a tick when the train comes out, and the message never appears twice. **PM decision:** how long to hold undelivered messages (longer helps people with a broken phone; shorter keeps less data), and what the sender sees in the meantime.

4. Group fan-out: one message, 46 deliveries

Everyday picture. Farhan's mother prints one wedding card design, then posts a copy to each relative. The design work happens once; the postage happens 46 times.

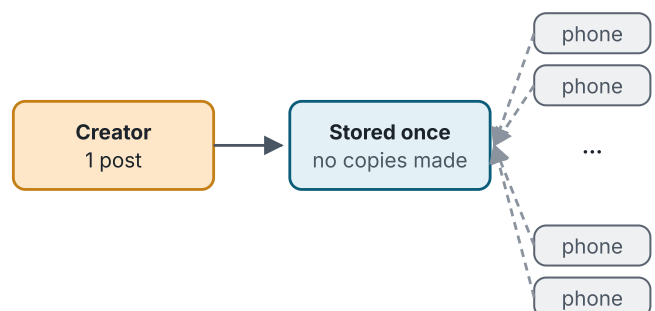
How it works. The aunt's phone sends the voice note to the server once. The server looks up the group's members and puts a copy into each member's delivery path, to their connection or to their offline queue. This is **fan-out on write**: the copying happens when the message is sent. For a WhatsApp Channel with one crore followers, copying every post into one crore inboxes would be wasteful, so channels use **fan-out on read**: the post is stored once, and each follower's phone fetches it when they open the channel (Figure 5.38). The maths for Farhan's group: 212 messages × 46 other members = **9,752 deliveries a day**, plus a delivered and a read receipt for each: about 29,000 small messages, from one family.

Group: fan-out on write



212 messages × 46 members = 9,752 deliveries a day, from one family

Channel: fan-out on read



1 crore followers: one stored post; only phones that open the channel fetch it

Figure 5.38 · Two ways to reach many people. Groups copy each message to every member, which suits a few hundred people who all read it. Channels store a post once and let followers pull it, which suits crores of followers who mostly don't. *Author's diagram; group and channel limits from WhatsApp (2022–23).*

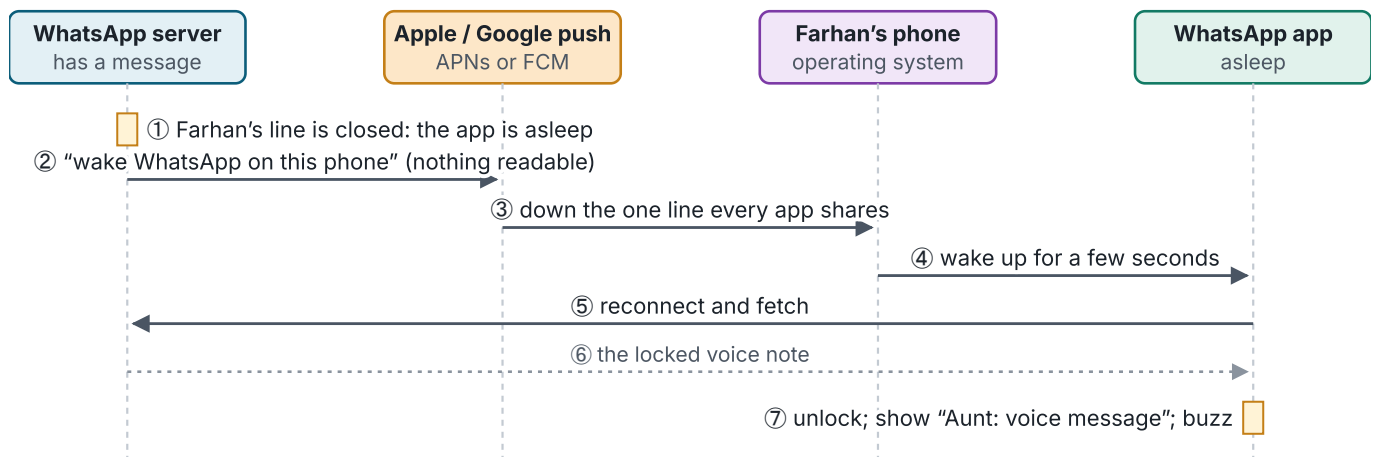
Real example. WhatsApp groups held 256 people for years. It raised the limit to 512 in 2022 and to 1,024 later that year, when it launched Communities, and it launched Channels in 2023 for one-to-many broadcasts. Telegram lets groups reach 2 lakh members. **PM decision:** the group size limit is both a cost and a product choice. Every extra

member multiplies deliveries and receipts, and very large groups become noisy and good at spreading rumours. The better design gives broadcasters a separate tool (Channels) instead of ever-bigger groups.

5. Push notifications: waking a sleeping app

Everyday picture. You can't hear a knock on your flat's door from the back room, so the society guard rings your intercom: "Parcel for you at the gate." The guard doesn't open the parcel; he only tells you it's there.

How it works. To save battery, iPhones and most Androids stop apps working in the background after a while, which closes WhatsApp's open line. So when a message arrives for a sleeping phone, WhatsApp's server sends a short request to the phone maker's own service: Apple's **APNs** or Google's **Firebase Cloud Messaging (FCM)**. Every phone keeps one connection open to that service, shared by all apps. It wakes WhatsApp for a few seconds; the app reconnects, fetches and unlocks the message, and shows "Aunt: Voice message" (Figure 5.39). Because chats are encrypted end to end, the push itself carries nothing readable, only a nudge or a locked message.



One shared line to Apple or Google for all apps saves battery; WhatsApp only asks it to knock.

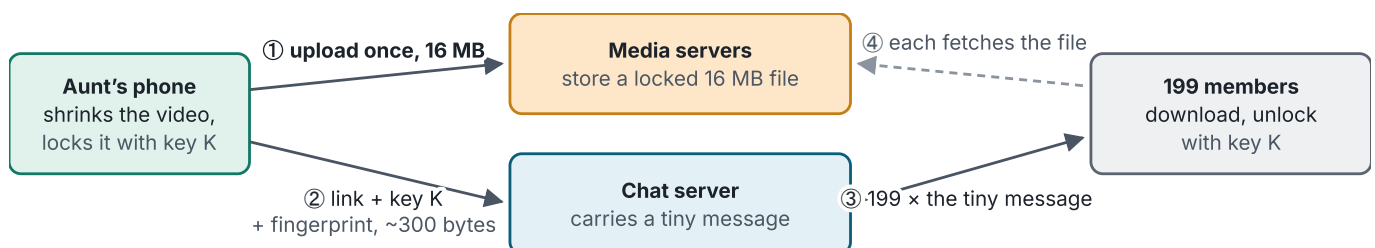
Figure 5.39 · How Farhan's pocket buzzes. One shared line to Apple or Google serves every app on the phone, which saves battery; WhatsApp only asks it to wake the app. *Simplified from Apple and Google developer documentation.*

Real example. Some phone brands popular in India shut background apps very aggressively to make the battery last longer. Messages then arrive only when the user opens the app, and people blame WhatsApp. Apps now show guides for turning off battery optimisation on specific brands. **PM decision:** what the lock screen shows (a message preview is convenient but anyone holding the phone can read it); how often to notify (big groups default to fewer alerts, and "mentions only" is a lifesaver); and testing on the cheap phones most users have, not only on the team's own.

6. Photos and videos: send a collection slip, not the parcel

Everyday picture. A heavy parcel stays at the post office; each person gets a slip to collect it.

How it works. The phone first shrinks the photo or video, then locks it with a fresh random key and uploads the locked file once to WhatsApp's media servers. The chat message carries only the link, the key and a fingerprint of the file (to check it arrived intact). Each member downloads the file and unlocks it with the key (Figure 5.40). The maths for a 16 MB wedding video sent to a 200-member group: sending it inside each message would mean $200 \times 16 \text{ MB} = 3.2 \text{ GB}$ leaving the aunt's phone; by link it is 16 MB, 200 times less. When someone forwards the same video, WhatsApp can reuse the file already stored instead of uploading it again.



Uploads from the aunt's phone:

file inside every message ($200 \times 16 \text{ MB}$) | **16 MB by link: 200 times less**

3.2 GB

Figure 5.40 · A 16 MB video to a 200-member group. The sender uploads once; the message is only a few hundred bytes. The servers store a locked file they can't open. *Illustrative sizes; the design follows WhatsApp's encryption white paper.*

Real example. WhatsApp's settings let users choose which media downloads automatically on mobile data, Wi-Fi or roaming, and since 2023 whether to send photos in HD. **PM decision:** quality against data. Many Indians are on a 1.5 GB-a-day pack; if a family group auto-downloads ten videos a day, that is 150 MB gone before lunch. Sensible defaults (photos yes, videos on Wi-Fi) matter more than any new feature.

7. Presence: "online" and "typing..."

Everyday picture. The light in a neighbour's window tells you they're home. Nobody writes down when it was on.

How it works. Presence updates are sent only to people who have your chat open right now, and are thrown away after. They don't need to be perfect, so they are handled the fast way: if one is lost, the next one fixes it. This is the "availability over consistency" choice made for a whole feature. **PM decision:** social cues against privacy. In 2022 WhatsApp let users hide "online" from everyone, after years of complaints from people who felt watched.

8. Forward limits: fighting rumours by design

Everyday picture. A post office that stamps chain letters "copied many times" and lets you send each one to only one address at a time.

How it works. WhatsApp can't read messages, so it can't check whether a forward is true. But the phones can count. Each message carries a **forward counter**, locked inside the encryption. After five hops the phones label it "Forwarded many times" and let it go to only one chat at a time. **Real example.** After rumours on WhatsApp led to mob killings in India in 2018, WhatsApp limited forwards to five chats in India that July, and worldwide in 2019. In April 2020, during Covid rumours, it added the one-chat limit for highly forwarded messages and later said this cut such forwards by 70%. **PM decision:** friction is a safety tool. It also slows useful forwards, like a school notice, so the limit applies only to messages that have already travelled far.

9. When a chat server dies

Everyday picture. The postman for your lane falls ill. The post office sends another one, the letters in the sorting room are still there, and you might get your post an hour late, but nothing is lost.

How it works. If a chat server crashes, the 20 lakh phones connected to it lose their line. Each phone reconnects through the load balancer to a healthy server, and the session directory is updated. Nothing waiting is lost, because offline queues are stored on several machines. Anything the phone sent but never saw a tick for, it simply sends again with the same message ID, and the ID stops it arriving twice. The danger is the **reconnect storm**: if 20 lakh phones all retry in the same second, they can knock over the next server too. So each phone waits a random time before retrying, and waits longer after each failure (Figure 5.41). Spread over 60 seconds, $20 \text{ lakh} \div 60 \approx 33,000$ reconnections a second: easy to absorb.

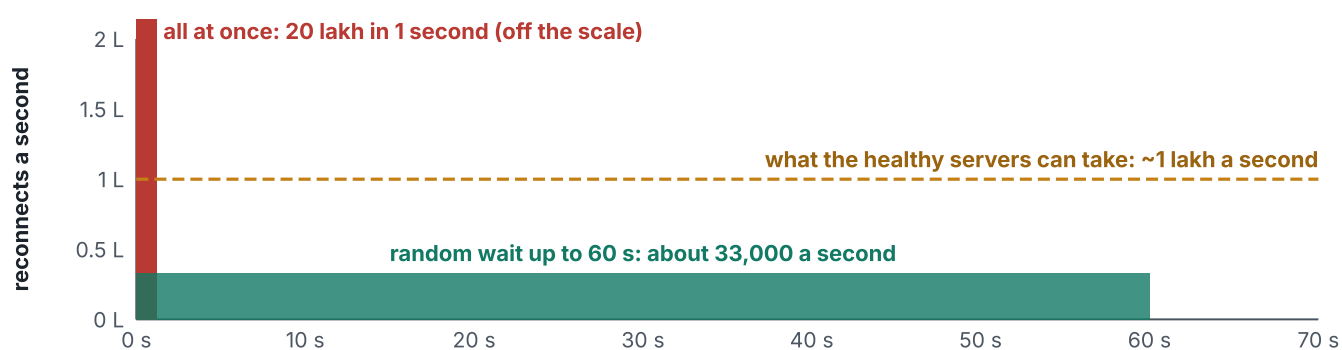


Figure 5.41 · Twenty lakh phones reconnecting. If all retry at once, the spike can take down the next server. A random wait of up to a minute turns the spike into a gentle slope. *Illustrative numbers; the random wait ("jitter") with growing gaps ("backoff") is standard practice.*

Real example. On 4 October 2021 WhatsApp, Instagram and Facebook all went down for about six hours. A routine network change at Meta cut its data centres off from the rest of the internet, and even its engineers' door badges stopped working. Telegram's founder said it gained about 7 crore users that day. **PM decision:** what users see while things are broken (the clock icon, an honest banner, never a false tick), how the company talks to users during an outage, and whether a burst of new sign-ups for a rival that day is a warning or a blip.

KEY IDEA · DIWALI MIDNIGHT, PLANNED IN ADVANCE

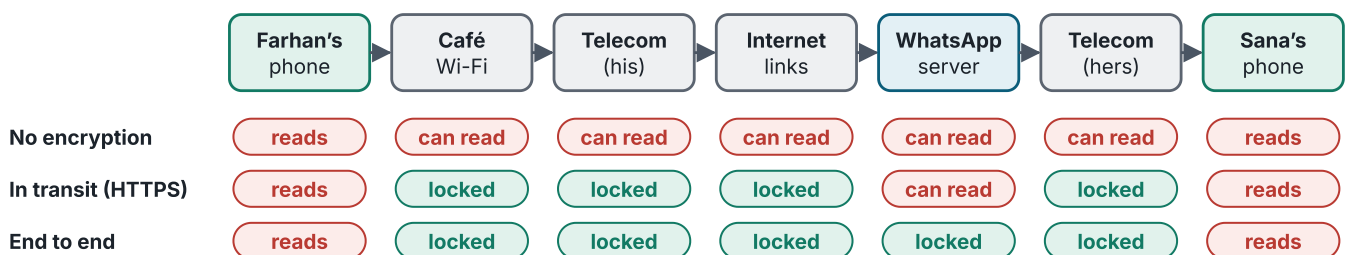
What happens at 12:00	What the system does	What the PM decides beforehand
Messages jump several times in minutes	Extra chat servers started in the evening; queues absorb the burst	Share the festival calendar; no risky releases that week
Everyone forwards the same greeting video	Stored once, reused on forward	Keep reuse on; default video downloads to Wi-Fi only
Some deliveries run late	Store and forward holds them; ticks stay honest	Show one tick rather than an error; never fake delivery
Notifications flood phones	Group alerts batched on the phone	"Mentions only" for big groups; no marketing pushes that night

Encryption: how a message stays private

Farhan is at a café in Hazratganj, on the café's free Wi-Fi. He types "Meet me at 5 pm" to his cousin Sana. The message passes through the café's router, his telecom company, the internet, WhatsApp's servers and Sana's telecom before it reaches her phone. Any of them could, in principle, look inside. Encryption decides who actually can. A PM needs this vocabulary: every privacy promise, every feature that needs to read messages, and every argument with a regulator comes back to it.

What can be seen on the way

Without any encryption, everyone along the path could read "Meet me at 5 pm". With encryption **in transit**, the lines are protected but WhatsApp's server unlocks and relocks the message, so the company could read it. With **end-to-end** encryption, only Farhan's and Sana's phones can read it; everyone in between, WhatsApp included, passes on a locked box (Figure 5.42). The ideas below explain how that works, from the smallest building block up.



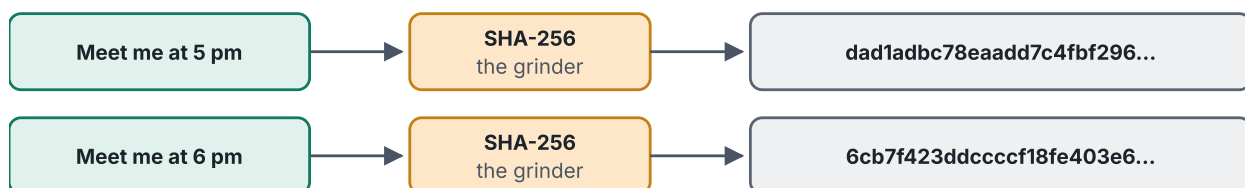
The ends must read the message. The question is who else can: everyone, the company, or nobody.

Figure 5.42 · Who could read Farhan's message. Each kind of protection moves the point where the message is readable. End to end, it is readable only at the two ends. *Author's diagram.*

1. Hashing: a fingerprint of any data

Everyday picture. Grinding masala. Whole spices go in and a powder comes out. The same spices always give the same powder, but nobody can turn the powder back into whole spices. A fingerprint works the same way: it identifies you, but no one can rebuild you from it.

How it works. A **hash function** turns any data, a word or a whole film, into a short code of fixed length. SHA-256, the most common, always gives 64 characters. Three rules make it useful: the same input always gives the same code; changing even one letter changes the whole code; and there is no way back from the code to the input (Figure 5.43).



Always 64 characters, for a word or a film.

one character changed → nothing in common

No way back: the code can be checked, never reversed.

Figure 5.43 · One character changed, a completely different fingerprint. The two messages differ only in "5" and "6"; their SHA-256 codes share nothing. *Real SHA-256 outputs, computed for this page (first 24 of 64 characters shown).*

Where it's used. Websites should never store your password, only its hash. At login they hash what you type and compare. WhatsApp puts the fingerprint of every photo and video inside the message, so the receiving phone can check that the file it downloaded wasn't changed on the way. **The maths of a leak.** Hashes can't be reversed, but they can be guessed. An 8-letter password in small letters has $26^8 \approx 2.1 \times 10^{11}$ possibilities. A gaming graphics card can

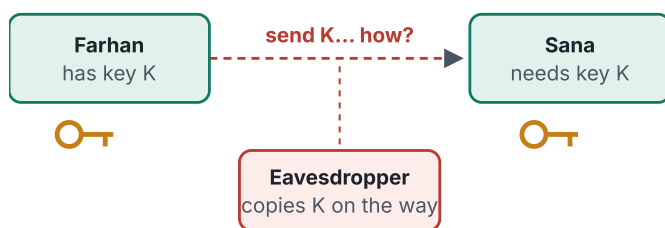
try about 10^{10} fast hashes a second, so it finds the password in about 21 seconds. With a hash designed to be slow (bcrypt, for example) it manages about 10^4 a second, and the same search takes about 8 months. Adding a random **salt** (extra characters, different for each user) before hashing means attackers must crack each person separately. **Real example.** In 2012 LinkedIn lost millions of password hashes that were fast and unsalted; most were cracked within days. **PM decision:** insist on slow, salted hashes; check new passwords against lists of leaked ones; prefer one-time passwords or passkeys where you can.

2. One shared key: symmetric encryption

Everyday picture. A steel almirah at home with two copies of its key. Whoever holds a copy can lock it and open it.

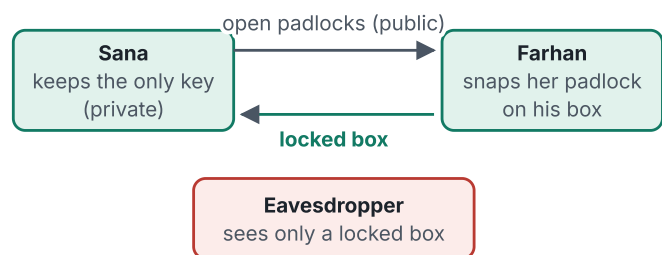
How it works. In **symmetric encryption** the same key locks and unlocks. The standard method, **AES**, scrambles data so thoroughly that without the key it looks like random noise. It is fast enough to lock video as it is recorded, which is why WhatsApp uses AES-256 for the content of every message and file. How strong is it? A 128-bit key has $2^{128} \approx 3.4 \times 10^{38}$ possible values. At a trillion guesses a second, trying them all would take about 10^{19} years, roughly 800 million times the age of the universe. **The catch:** Sana needs a copy of the key. If Farhan sends the key over WhatsApp, anyone watching gets the key too. This is the **key-sharing problem**, and the next three ideas exist to solve it (Figure 5.44).

One shared key (AES)



Fast, but the key must travel, and a **copied key opens everything**.

Open padlocks (public and private keys)



Padlocks can be handed out freely: only **Sana can open what they lock**.

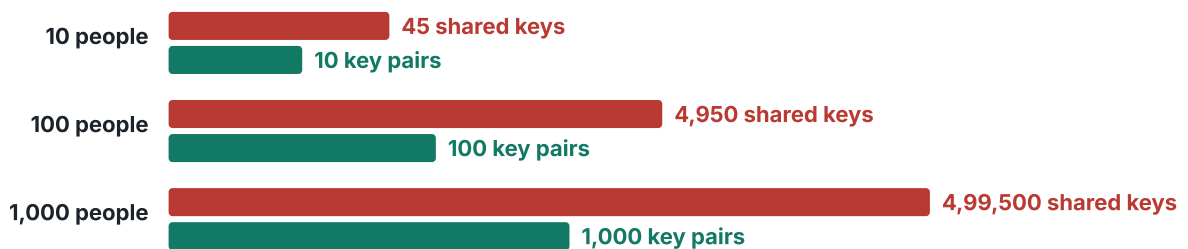
Figure 5.44 · Two kinds of lock. With one shared key, the hard part is getting the key to the other person safely. With open padlocks, Sana can hand them out freely, because only she can open them. *Author's analogy diagram.*

3. Public and private keys: open padlocks anyone can close

Everyday picture. Sana gives out open padlocks to anyone who asks. Farhan puts his note in a box and snaps her padlock shut. Now even Farhan can't open the box; only Sana, who kept the only key, can.

How it works. Each person has a **key pair**. The **public key** is the open padlock: shared freely, stored in a directory. The **private key** is the only key that opens it, and it never leaves the owner's phone. Anything locked with Sana's public key opens only with her private key. Key pairs are slow, so in practice they lock something small, a fresh AES key, and the AES key locks the message. **The maths of scale.** If 1,000 people each want a private line to every other person using shared keys, they need $1,000 \times 999 \div 2 = 4,99,500$ secret keys, each handed over safely. With key pairs, each person needs just one: 1,000 public keys in a directory (Figure 5.45).

Keys needed for everyone to talk privately in pairs (bar length on a log scale)



$n \times (n - 1) \div 2$ against n : for WhatsApp's 3 bn users, shared keys would be about 4.5×10^{18} .

Figure 5.45 · Keys needed for private pairs. Shared keys grow with the square of the group; key pairs grow in step with the number of people. That is what makes encryption possible for three billion users. *Simple arithmetic: $n \times (n - 1) \div 2$ shared keys against n key pairs.*

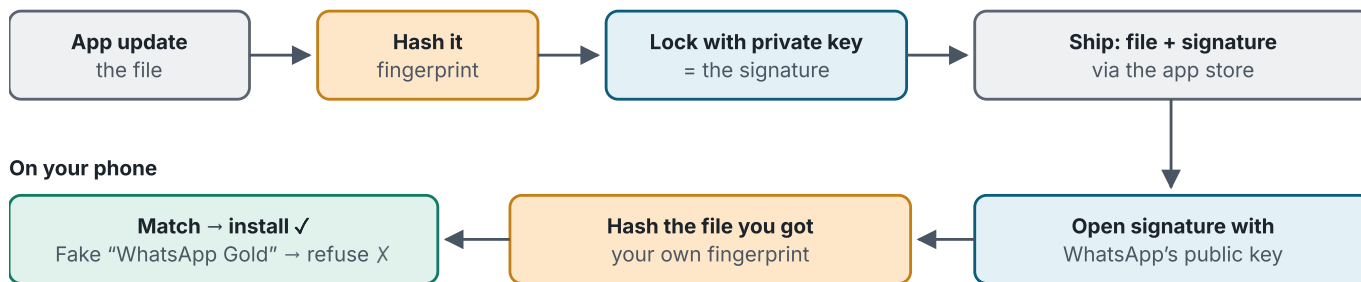
Real example. The padlock in your browser's address bar when you open your bank's website: the bank's server sends its public key, and your browser uses it to set up the connection. **PM decision:** someone must run the directory of public keys; for WhatsApp, it is WhatsApp's own server. That raises the obvious question, what if the directory lies? The answer is the security code: a check that both phones hold the same padlocks.

4. Digital signatures: a seal only you can make

Everyday picture. A king's wax seal, pressed with his ring. Anyone can recognise it; only the king can make it; and a broken seal shows the letter was opened.

How it works. To sign, the sender hashes the message and locks the hash with their *private* key. To check, anyone opens it with the sender's *public* key and compares it with their own hash of the message. If they match, the message came from the key's owner and wasn't changed by even one character (Figure 5.46).

At WhatsApp



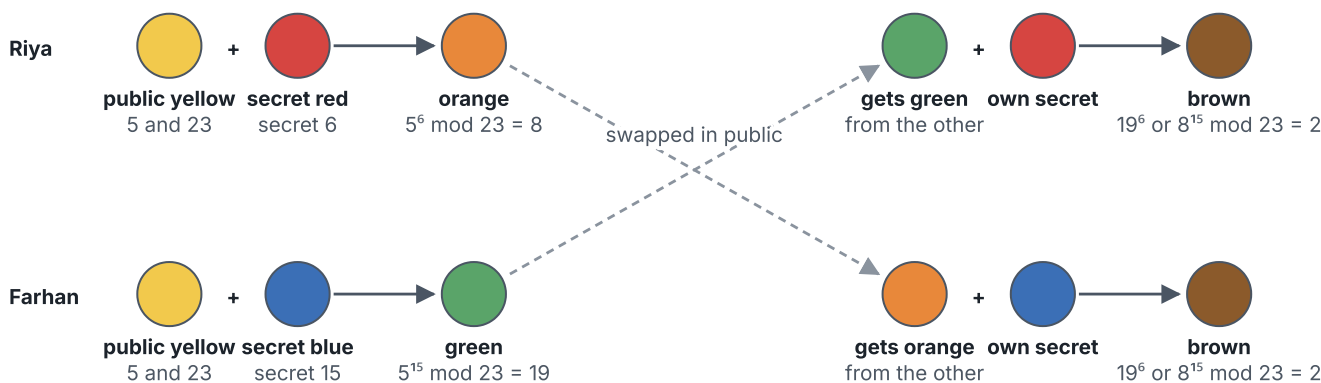
Only the private key can make the signature; anyone with the public key can check it.

Figure 5.46 · Signing and checking an app update. Only WhatsApp's private key can make the signature; every phone can check it with the public key. A fake app fails the check. *Simplified from how Android and iOS verify app signatures.*

Real examples. Phones install an app update only if it is signed with the same key as the installed app, so a fake "WhatsApp Gold" file sent on a group can't replace the real app. Aadhaar eSign and documents in DigiLocker carry the issuer's digital signature. **PM decision:** guard the signing key as the company's most valuable secret, and never train users to install app files from links; that is how many Indian "bank KYC" and "PM scheme" scams spread malware.

5. Agreeing a secret in public: key exchange

Everyday picture. Mixing paint (Figure 5.47). Riya and Farhan both start with the same yellow, which anyone can see. Riya secretly adds red and gets orange; Farhan secretly adds blue and gets green. They swap orange and green in public. Riya adds her red to Farhan's green; Farhan adds his blue to Riya's orange. Both end up with the same brown. Someone who saw yellow, orange and green can't unmix them to find the brown.



An eavesdropper sees yellow, orange and green, but can't unmix them to make the brown.

Figure 5.47 · Agreeing a secret colour in public. The mixtures that travel in the open are useless to an eavesdropper; the final brown is made only on the two phones. *The paint analogy for Diffie-Hellman key exchange; numbers in the text.*

How it works, with small numbers. This is **Diffie-Hellman key exchange** (1976). "mod" means the remainder after dividing, like clock arithmetic. Public numbers: 23 and 5. Riya picks a secret, 6, and sends $5^6 \bmod 23 = 8$. Farhan picks a secret, 15, and sends $5^{15} \bmod 23 = 19$. Riya computes $19^6 \bmod 23 = 2$; Farhan computes $8^{15} \bmod 23 = 2$. Both have the secret 2, and it never travelled. With numbers hundreds of digits long, working backwards from 8 and 19 is practically impossible. **Real example.** Every HTTPS connection and every new WhatsApp chat starts this way; WhatsApp uses a modern version called Curve25519. **PM decision:** because fresh secrets are cheap, systems can make new ones often, which limits the damage when one leaks.

6. In transit or end to end

Everyday picture. A courier company that opens your parcel at its warehouse, checks it and seals it again, against a locked box that only the receiver has a key for. In the second case the courier sees only the address label.

How it works. With encryption **in transit** (TLS, the “S” in HTTPS), the message is locked between your phone and the company’s server, unlocked there, and locked again for the receiver. The company can read and store it. With **end-to-end encryption**, the message is locked on your phone and only the receiver’s phone can open it; the server passes on a box it can’t open (Figure 5.48).

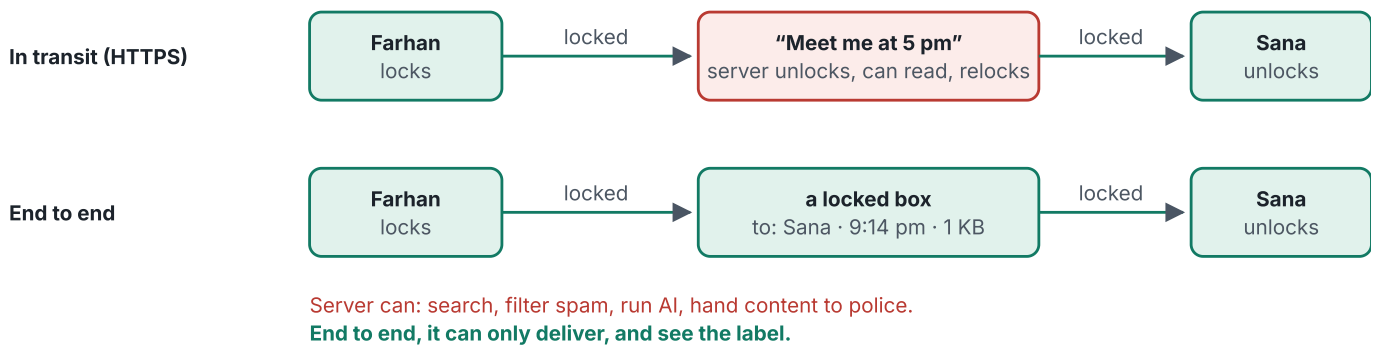


Figure 5.48 · What the company’s server sees. In transit, the server holds the readable message for a moment, which lets it search, filter spam and run AI on it. End to end, it holds only the locked box and the address label. *Author’s diagram.*

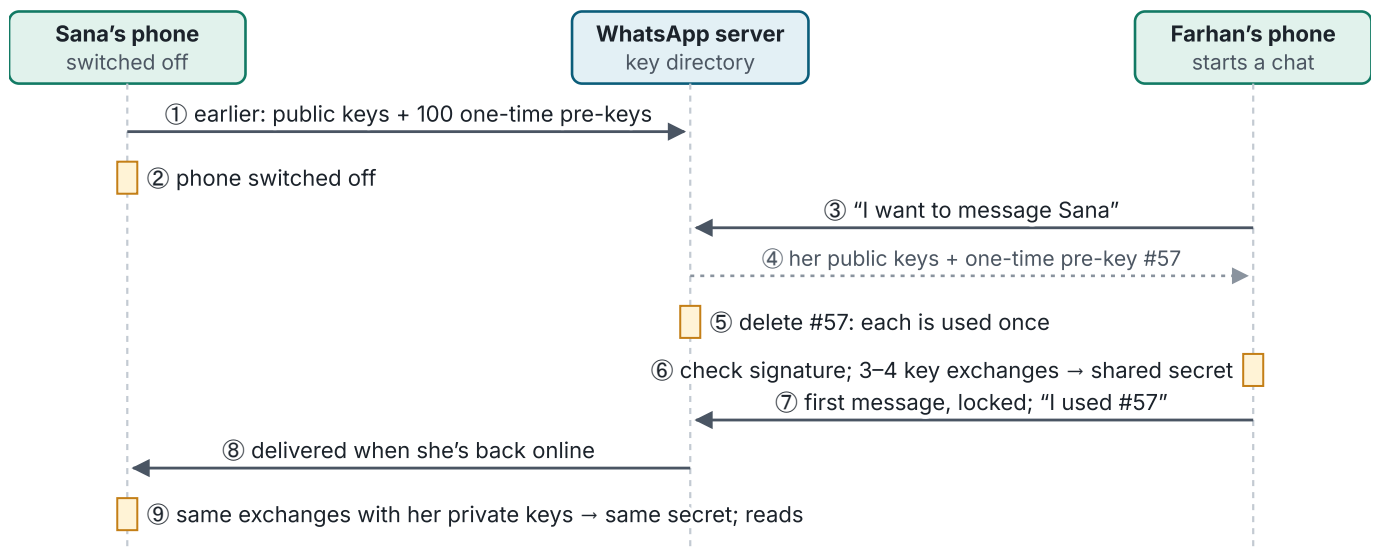
Real examples. Gmail is encrypted in transit, which is how it can filter spam and suggest replies. Telegram’s normal chats are too: stored on its servers so they sync to any device, and readable by the company. WhatsApp, Signal and iMessage are end to end, and Messenger became end to end by default in December 2023. **PM decision:** end to end is a promise users can check and journalists will test. Every feature that needs the server to read messages must be dropped, moved onto the phone, or built another way.

7. How WhatsApp’s end-to-end encryption works

WhatsApp uses the **Signal protocol**, built by the makers of the Signal app and switched on for every WhatsApp chat in April 2016. It combines the ideas above in five steps.

(a) Keys are made on the phone. When you install WhatsApp, your phone creates an **identity key pair** (your long-term padlock and key), a **signed pre-key**, and a batch of **one-time pre-keys**. The public halves go to WhatsApp’s server. The private halves never leave your phone.

(b) Starting a chat with someone who is offline. Key exchange normally needs both people present, like the paint swap. But Sana’s phone might be switched off. So her phone has already left a stack of open padlocks, the one-time pre-keys, at the “post office”. Farhan’s phone collects one, does the key exchange with it on its own, and sends the first message along with what Sana needs to make the same secret when she comes online (Figure 5.49). The Signal protocol calls this X3DH.

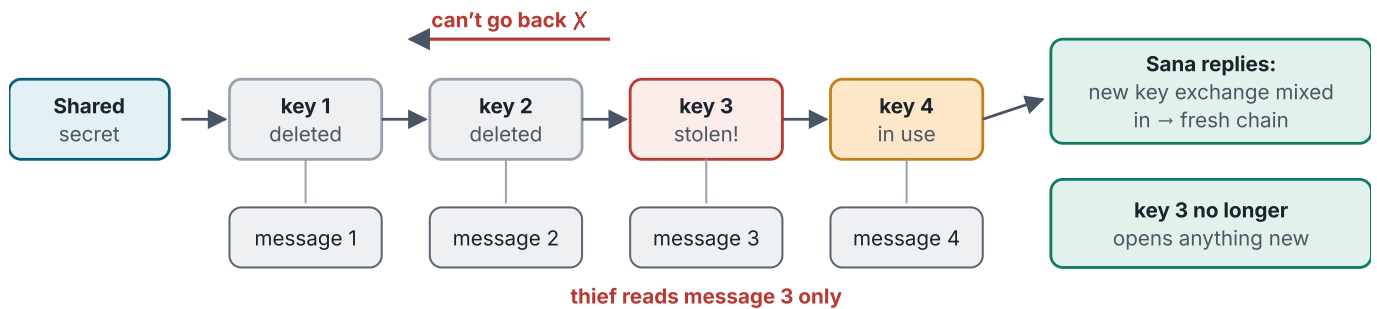


Pre-keys let a conversation start safely even when the other phone is off.

Figure 5.49 · Starting a private chat with a phone that is switched off. Sana’s pre-keys wait on the server like a stack of open padlocks; each is used for one new conversation, then thrown away. *Simplified from the WhatsApp Encryption Overview (white paper) and the Signal protocol’s X3DH specification.*

(c) A new key for every message. From that first secret, each message gets its own key, made by turning a **ratchet**: a one-way step, like a car jack that only turns one way. Old keys are deleted as soon as they are used. So if a thief copies today’s key, yesterday’s messages stay locked: this is **forward secrecy**. And each time the other person replies, a

fresh key exchange is mixed in, so a stolen key stops working after a round of messages (Figure 5.50). The everyday picture is a pad of one-time codes: each page used once, torn off and burned.

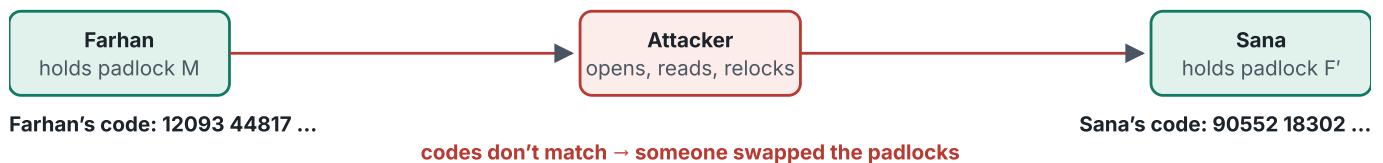


Each step is one-way, like a car jack: forward secrecy (the past stays locked) and self-healing (the future re-locks).

Figure 5.50 · The ratchet: one key per message, old keys destroyed. A thief who steals key 3 can read message 3 and nothing before it; when Sana replies, a new exchange locks the thief out again. *Simplified from the Signal protocol's Double Ratchet specification.*

(d) Checking nobody is in the middle. The weak point is the directory. If an attacker, or a dishonest server, gave Farhan a fake padlock that looked like Sana's, the attacker could open messages, read them and relock them with Sana's real padlock. Neither would notice. So WhatsApp shows each pair a **security code**: 60 digits, or a QR code, made from both people's identity keys. If the code on Farhan's phone matches the code on Sana's, nobody is in the middle (Figure 5.51). WhatsApp also warns "security code changed" when a contact gets a new phone, and since 2023 it has checked keys automatically against a public log (key transparency).

With someone in the middle



Nobody in the middle

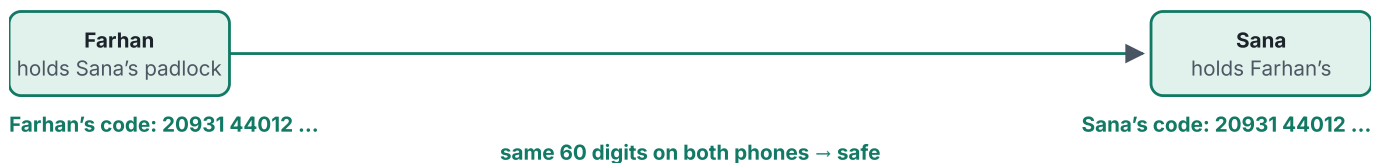


Figure 5.51 · Why the security code matters. A swapped padlock lets someone read and relock messages unnoticed. Comparing the code on both phones exposes the swap, because the codes are made from the padlocks each phone actually holds. *Author's diagram, after the WhatsApp Encryption Overview.*

(e) Groups, linked devices and backups. **Groups** use **sender keys**. Each member makes a group key and sends it once, privately, to every other member. After that, each message is locked once and the server copies it to every member, as for any group message. In Farhan's 47-member group, setting up keys takes $47 \times 46 = 2,162$ private key messages; after that, each message is locked once, not 46 times. When someone leaves, keys are changed so they can't read new messages. **Linked devices** (a laptop or a tablet) each get their own identity key, and messages are locked separately for each device: Sana with a phone and a laptop receives two copies. **Backups** go to Google Drive or iCloud and, by default, are not end to end. Since 2021, users can lock their backup with a password or a 64-digit key. The key is kept in a tamper-proof hardware vault that allows only a few wrong guesses. Lose the password and the 64-digit key, and the backup is gone for good; nobody can recover it.

8. What end-to-end encryption doesn't hide

End to end protects what you say, not the fact that you said it (Figure 5.52). WhatsApp's servers must know who is sending to whom, and when, or they couldn't deliver anything. They also see your phone number, IP address, phone model and who is in which group. That information about messages is called **metadata**, and it can reveal a lot: a call to an oncologist at 11 pm says something even if nobody hears it. Beyond metadata: anyone holding your unlocked phone can read everything; the person you send to can screenshot or forward it; a backup that isn't encrypted end to end can be read by whoever can open that cloud account; and when you report a chat, your phone sends the last few messages to WhatsApp for review. Spyware on the phone itself reads messages before they are locked: in 2019 WhatsApp found that the Pegasus spyware had targeted about 1,400 users, including Indian journalists and activists, and sued its maker, NSO Group; a US jury ordered NSO to pay about \$168 mn in 2025.

Locked: only the two phones can read

- ✓ Message text
- ✓ Photos, videos, documents
- ✓ Voice notes and calls
- ✓ Status updates
- ✓ Backups, if the user turns on encryption

Not locked by end to end

- Who messages whom, and when (metadata)
- How often; your phone number, IP, phone model
- Who is in which group
- Anything on an unlocked or hacked phone
- Screenshots and forwards by the receiver
- Backups left unencrypted; reported messages

Figure 5.52 · What is locked, and what is not. The content is locked; the envelope (who, when, how often) is not, and the two ends are only as safe as the phones. *WhatsApp Encryption Overview and privacy policy; author's summary.*

9. The PM trade-offs

Feature or duty	Why end to end makes it hard	What WhatsApp does instead
Searching old chats	The server can't read, so it can't search	Search runs on the phone
Spam and scams	No scanning of message text	Behaviour signals (a new number messaging hundreds of strangers), user reports, rate limits, forward counts
AI summaries	The server can't see what to summarise	Opt-in "Private Processing" (announced 2025): the messages are handled in sealed hardware that even Meta says it can't look into
A lost phone	No readable copy exists anywhere	Backups, encrypted if the user chooses; a clear warning at setup
Misinformation	Can't see what is spreading	Forward limits and labels; fact-check tip lines on WhatsApp itself
Police requests	No message content to hand over	Account details and metadata, under legal process

The traceability debate in India. Rule 4(2) of India's IT Rules 2021 says a large messaging service must be able to identify the *first originator* of a message when a court or the government orders it, for serious crimes. WhatsApp argues that the only way to do this is to attach a fingerprint of the sender to every message sent by every Indian, just in case one is ever asked about, which would break the promise of end-to-end encryption for everyone. It challenged the rule in the Delhi High Court in May 2021, and in April 2024 its lawyer told the court that WhatsApp would leave India rather than break encryption. The court last heard the case in August 2024 and had not ruled as of September 2026. The government's side is real too: a forwarded rumour that leads to violence has a source, and investigators want to find it. **PM decision:** in any market, know which promise you would refuse to break, and design the product (forward limits, labels, reporting) so that the company can do as much as possible without breaking it.

IN THE INTERVIEW · HOW ENCRYPTION COMES UP

- **"Should WhatsApp add AI summaries?"** Only on the phone or in sealed processing, opt-in, because the server can't read chats. Say what you give up: slower, fewer phones supported.
- **"How would you reduce scams on an encrypted app?"** Use what the server can see: new accounts, messages to many strangers, reports, forwards. Warn users when a stranger sends a link or asks for money.
- **"A government asks you to trace a message."** Explain the trade-off: tracing one message means fingerprinting all of them. Offer metadata under legal process, forward limits and labels instead.
- **"What is end-to-end encryption, simply?"** The message is locked on the sender's phone and only the receiver's phone has the key; the company carries a box it can't open.

Connect the dots

Four ideas ran through the cases, the products and the technology. A PM at a social or messaging company meets all four in the same week, often in the same family WhatsApp group.

Thread	In the cases	In the products	In the technology
Density before size	The jobs app works only with ~450 households in one suburb; crores of thin sign-ups would still leave Savita empty-handed	Threads borrowed Instagram's network; Snap's streaks keep a small circle active daily	Every group member multiplies deliveries and receipts; broadcasts move to Channels
Trust is the product	"A worker never pays"; WhatsApp's problem was belief about privacy, not missing features	No ads in chats; Community Notes need agreement across sides; the ad auction penalises reported ads	End to end, security codes, honest ticks; the traceability debate
Who pays, and for what	Households pay per hire; creators earning is the goal, GMV only an input	Bid × chance + quality; WhatsApp charges businesses per message; Snap adds subscriptions	Media by link saves users' data packs; heartbeats save their batteries
People keep two apps	New groups start on Telegram while family groups stay; workers keep Apna too	WhatsApp for people you know, Telegram for strangers; Threads beside X	Push notifications and battery rules decide which app reaches you first

BACK TO LUCKNOW · A SKETCH ANSWER FOR THE FAMILY-GROUP QUESTION

Why the move failed. Everyone had to move together, and they didn't. If each of 47 relatives has a 90% chance of joining, the chance that all do is 0.9^{47} , under 1%. One grandmother who stays on WhatsApp keeps the wedding plans there. The group also had years of photos and voice notes on WhatsApp, and the trigger was a false rumour: WhatsApp wasn't charging for groups. Telegram offered this family nothing they needed. Even Telegram's January 2021 feature for importing WhatsApp chat history lowered the cost of losing old messages, not the cost of moving people.

What it would take. A rival can't win the old group; it must win a new job that WhatsApp does badly for this family, and grow from there. Examples: a 300-guest wedding group where most guests are strangers and don't want to share phone numbers, or a family photo album that keeps full-quality videos for years. Each is a new group with low switching costs. Over time, if enough of the family's new activity lives elsewhere, the old group quietens.

What WhatsApp should watch. Goal (North Star): weekly active senders, by age group. **Safety checks (guardrails):** where new groups are created; messages per family group; spam reports. **Trade-off:** opening up to strangers (usernames, bigger groups) wins new jobs but weakens the "private, people you know" promise that holds the old ones.

KEY IDEA · NETWORKS ARE WON ONE GROUP AT A TIME

Social and messaging products are made of small networks: a family, a college batch, one suburb's households. Each is won or lost as a whole. When you design, measure or defend such a product, ask first: which small group is this for, what makes it worth coming back to, and what would it take for all of them to move?